

ON THE VC-DIMENSION OF ERDŐS–RÉNYI RANDOM GRAPHS

XIZHI LIU AND HENG LI

ABSTRACT. Let $D_n^\circ(p)$ and $D_n^{\text{cl}}(p)$ be the VC-dimensions of the open- and closed-neighborhood systems of $G(n, p)$. We determine their asymptotic behavior at symmetry, for every fixed asymmetric parameter, and throughout the crossover $p_n \rightarrow 1/2$. At $p = 1/2$, both dimensions are concentrated on the two integers adjacent to $\log_2(n/\log n)$, and the number of shattered sets at the critical level has a Poisson limit. For fixed $p \neq 1/2$, we locate both dimensions within a bounded window and identify the rare Hamming layers that determine the second-order term; on critical sequences in this scale, the corresponding count again converges to a Poisson law. For an arbitrary sequence $p_n \rightarrow 1/2$, we introduce a moving contact scale $B_k(\theta_n)$ and its last feasible lattice point K_n . Both dimensions then lie in $\{K_n, K_n + 1\}$ with high probability, are asymptotically equal, and satisfy

$$\mathbb{P}(D_n^\circ(p_n) = D_n^{\text{cl}}(p_n) = K_n + 1) = 1 - \exp\{-\exp(H_{n,k} - \Phi_{n,k})\} + o(1), \quad k = K_n + 1,$$

where $H_{n,k} = \log \binom{n}{k}$ and $\Phi_{n,k}$ is the exact Poissonized occupancy exponent. A variational formula gives a uniform deterministic center, while a tilted contact partition resolves the microscopic Poisson window. The proof combines relative de-Poissonization with a shared/private Poisson interpolation that is uniform over all overlap sizes.

1. INTRODUCTION

Let $\mathcal{F} \subseteq 2^V$ be a set system. A set $S \subseteq V$ is *shattered* by $\mathcal{F}$ if

$$\{F \cap S : F \in \mathcal{F}\} = 2^S.$$

The VC-dimension of $\mathcal{F}$ is the maximum size of a shattered set. Introduced by Vapnik and Chervonenkis [10], it is a basic measure of combinatorial complexity; the Sauer–Shelah lemma [8, 9] gives the fundamental growth bound for set systems of bounded VC-dimension.

A graph $G = (V, E)$ produces the open- and closed-neighborhood systems

$$\mathcal{N}^\circ(G) := \{N_G(v) : v \in V\}, \quad \mathcal{N}^{\text{cl}}(G) := \{N_G[v] : v \in V\}.$$

We set

$$D^\circ(G) := \text{VCdim}(\mathcal{N}^\circ(G)), \quad D^{\text{cl}}(G) := \text{VCdim}(\mathcal{N}^{\text{cl}}(G)),$$

and, for the random graph,

$$D_n^\circ(p) := D^\circ(G(n, p)), \quad D_n^{\text{cl}}(p) := D^{\text{cl}}(G(n, p)).$$

When a statement applies to both conventions, $D_n(p)$ denotes either one.

Graph-defined set systems were studied systematically by Kranakis, Krizanc, Ruf, Urrutia, and Woeginger [6]; see also recent work on computation [3] and pseudorandom graphs [7]. Anthony, Brightwell, and Cooper [1] studied sparse thresholds for prescribed fixed neighborhood VC-dimension. Random binary function classes and random uniform hypergraphs provide closely related shattering models [11, 12], while random covering arrays lead to the same fixed-candidate

Date: August 23, 2026.

2020 Mathematics Subject Classification. 05C80, 60C05, 60F05, 05D40.

Key words and phrases. VC-dimension, random graph, shattering, occupancy, Poisson approximation, symmetry crossover.

coupon-coverage problem without the symmetry constraints of an adjacency matrix [4]. The present problem is different in two ways: the dimension grows with n , and candidate sets share edges, creating dependencies at every overlap size. To the best of our knowledge, the results below are the first sharp asymptotic laws for the growing neighborhood VC-dimension of dense Erdős–Rényi graphs, including critical lattice probabilities and the crossover through $p = 1/2$.

The common mechanism is occupancy. Fix a candidate set $S = \{x_1, \dots, x_k\}$. Each vertex outside S produces the binary trace $(\mathbf{1}_{vx_i \in E})_{i=1}^k$. Thus S is externally shattered exactly when every one of the 2^k binary patterns occurs. When $p = 1/2$, the patterns are equiprobable and ordinary coupon collection governs the threshold. For fixed $p \neq 1/2$, their probabilities vary geometrically across Hamming layers and finitely many rare layers determine the second-order term. For $p_n \rightarrow 1/2$, the dominant layer moves continuously from the center of the cube toward its rare side. The purpose of this paper is to establish the sharp laws at the two endpoints and to identify the entire transition between them.

Throughout, $G(n, p)$ is the random graph on $[n]$ with independent edge probability p . Unless a base is displayed, logarithms are natural. The phrase “with high probability” means with probability tending to one, and $O_{\mathbb{P}}(1)$ denotes boundedness in probability.

We first state the endpoint laws. For the symmetric model, set

$$J_n = \left\lfloor \log_2 \frac{n}{\log n} \right\rfloor.$$

Theorem 1.1. *Let $G \sim G(n, 1/2)$, and let D_n denote either neighborhood VC-dimension. Then $D_n \in \{J_n, J_n + 1\}$ with high probability. At $k = J_n + 1$, put*

$$M = 2^k, \quad u = \frac{n - k}{M}, \quad \lambda = Me^{-u}, \quad H_k = \log \binom{n}{k}.$$

If $\lambda - H_k \rightarrow +\infty$, then $D_n = J_n$ with high probability; if $H_k - \lambda \rightarrow +\infty$, then $D_n = J_n + 1$ with high probability; and if $H_k - \lambda \rightarrow \tau \in \mathbb{R}$, then

$$\mathbb{P}(D_n = J_n + 1) \rightarrow 1 - e^{-e^\tau}.$$

In the critical window, $H_k - \lambda \rightarrow \tau$ is equivalently

$$u = \log M - \log H_k + \frac{\tau}{H_k} + o(H_k^{-1}).$$

In particular, the central scale $\lambda/H_k \rightarrow 1$ has the expansion

$$u = \log n - 3 \log \log n + \log \log 2 + o(1).$$

For a fixed asymmetric parameter, let

$$q = \min\{p, 1 - p\}, \quad \rho = \frac{1 - q}{q}, \quad c_p = \max_{r \geq 3} \frac{r - 2}{\rho^r}.$$

Theorem 1.2. *Fix $p \in (0, 1) \setminus \{1/2\}$. For either neighborhood convention,*

$$D_n(p) = \log_{1/q} n - \log_{1/q} \log \log n - \log_{1/q} c_p + O_{\mathbb{P}}(1).$$

At an asymmetric critical lattice point put

$$a = (n - k)q^k, \quad H_k = \log \binom{n}{k}, \quad \Phi_k = \sum_{r=0}^k \binom{k}{r} F(a\rho^r), \quad F(x) = -\log(1 - e^{-x}).$$

Let Y_k be the number of externally shattered k -sets.

Theorem 1.3. *Suppose $a = \Theta(\log \log n)$ and $H_k - \Phi_k \rightarrow \tau \in \mathbb{R}$. Then*

$$Y_k \Rightarrow \text{Poisson}(e^\tau),$$

and, for either neighborhood convention,

$$\mathbb{P}(D_n(p) = k) \rightarrow 1 - e^{-e^\tau}, \quad \mathbb{P}(D_n(p) = k - 1) \rightarrow e^{-e^\tau}.$$

Moreover, the critical equation itself rules out the upper neighbor: for some $\delta_p > 0$, the corresponding exponent satisfies $\Phi_{k+1} \geq (\log n)^{2+\delta_p}$.

The constant c_p is analyzed in [Theorem 4.6](#): its maximizers form one Hamming layer or two adjacent layers. We next state the moving-parameter law that connects this rare-layer mechanism to the symmetric coupon collector.

Let $p_n \rightarrow 1/2$ and define

$$q_n := \min\{p_n, 1 - p_n\}, \quad \rho_n := \frac{1 - q_n}{q_n} = e^{\theta_n}, \quad \alpha_n := \log \frac{1}{q_n}, \quad a_{n,k} := (n - k)q_n^k.$$

In the moving-parameter results below, we suppress the argument and write D_n° and D_n^{cl} for $D_n^\circ(p_n)$ and $D_n^{\text{cl}}(p_n)$, respectively. The moving contact scale is

$$B_k(\theta) := \max_{0 \leq r \leq k} e^{-\theta r} \log \binom{k}{r},$$

and the moving lattice point is

$$(1) \quad K_n := \max\{0 \leq k < n : a_{n,k} \geq B_k(\theta_n)\}.$$

At $k = K_n + 1$, let

$$H_{n,k} := \log \binom{n}{k}, \quad \Phi_{n,k} := \sum_{r=0}^k \binom{k}{r} F(a_{n,k} \rho_n^r), \quad \lambda_n := e^{H_{n,k} - \Phi_{n,k}}.$$

Theorem 1.4. *For every sequence $p_n \rightarrow 1/2$,*

$$(2) \quad \mathbb{P}(D_n^\circ = D_n^{\text{cl}} = K_n + 1) = 1 - e^{-\lambda_n} + o(1),$$

$$(3) \quad \mathbb{P}(D_n^\circ = D_n^{\text{cl}} = K_n) = e^{-\lambda_n} + o(1).$$

In particular, both dimensions are supported on $\{K_n, K_n + 1\}$ with high probability and are equal with probability tending to one.

The corresponding counts satisfy a Poisson law, stated precisely in [Theorem 8.3](#). In particular, $H_{n,k} - \Phi_{n,k} \rightarrow \tau$ gives mean e^τ and the lattice probability $1 - e^{-e^\tau}$.

Define the binary entropy and crossover function

$$(4) \quad h(x) = -x \log x - (1 - x) \log(1 - x), \quad \mathcal{C}(A) := \max_{0 \leq x \leq 1} h(x) e^{-Ax}.$$

Theorem 1.5. *Let $L_n = \log n$, $A_n = \theta_n L_n / \alpha_n$, and*

$$(5) \quad m_n := \frac{L_n - \log(L_n / \alpha_n) - \log \mathcal{C}(A_n)}{\alpha_n}.$$

Then $K_n = m_n + O(1)$ and

$$D_n^\circ = m_n + O_{\mathbb{P}}(1), \quad D_n^{\text{cl}} = m_n + O_{\mathbb{P}}(1).$$

Corollary 1.6 (Three macroscopic regimes). *With the preceding notation:*

(i) if $A_n \rightarrow 0$, then

$$m_n = \frac{\log n - \log \log n}{\log 2} - \frac{A_n^2}{8} + o(A_n^2 + 1);$$

(ii) if $A_n \rightarrow A \in (0, \infty)$, then

$$m_n = \frac{\log n - \log \log n}{\log 2} + \chi(A) + o(1), \quad \chi(A) = \frac{\log \log 2 - \log \mathcal{C}(A) - A/2}{\log 2};$$

(iii) if $A_n \rightarrow \infty$, then

$$m_n = \frac{\log n + \log \theta_n - \log \log(\theta_n \log n / \alpha_n) + 1 + o(1)}{\alpha_n}.$$

The global theorem contains a second, microscopic transition. At $k = K_n + 1$, choose $r_* \in \arg \max_{0 \leq r \leq k} \rho_n^{-r} \log \binom{k}{r}$, put $L_* = \log \binom{k}{r_*}$, and define

$$Z_{k, \theta_n}(T) = \sum_{r=0}^k \binom{k}{r} e^{-B_k(\theta_n) \rho_n^r} e^{T \rho_n^{r-r_*}}.$$

There is a unique $\hat{T}_n \in (0, 3 \log k)$ such that $Z_{k, \theta_n}(\hat{T}_n) = H_{n,k}$. The actual tilt is

$$(6) \quad T_n := L_* \left(1 - \frac{a_{n,k}}{B_k(\theta_n)} \right).$$

Theorem 1.7. For fixed $\tau \in \mathbb{R}$,

$$H_{n,k} - \Phi_{n,k} \rightarrow \tau \iff H_{n,k}(\hat{T}_n - T_n) \rightarrow \tau.$$

Thus, with k and θ_n fixed in this deterministic calibration, the nontrivial Poisson window has width $H_{n,k}^{-1}$ in the tilt, $(H_{n,k} \rho_n^{r_*})^{-1}$ in $a_{n,k}$, and $(H_{n,k} L_*)^{-1}$ in the relative contact deficit.

When $\theta_n k$ remains bounded, [Proposition 10.5](#) reduces the tilted partition to an explicit Gaussian prefactor.

The exact exponent also recovers an explicit perturbative window. As shown in [Theorems 9.2](#) and [9.3](#), the symmetric lattice pair persists when $\varepsilon_n^2 (\log n)^3 = o(\log \log n)$, and the first change in its limiting probability occurs on the scale $\varepsilon_n (\log n)^{5/2} = \Theta(1)$.

The proof has a common probabilistic core and three parameter-specific analyses. Poissonization turns each fixed-set coverage event into an exact product. Relative de-Poissonization is essential because relevant probabilities can be $\exp\{-\Theta((\log n)^2)\}$. A shared/private Poisson interpolation compares simultaneous coverage of several cylinder partitions with the product of their marginals. At symmetry this gives a uniform cylinder estimate; at fixed asymmetry an exponent gap suppresses all overlapping factorial-moment contributions; in the global crossover a new contact scale yields a uniform $o(1)$ correction for every overlap size.

[Section 2](#) develops the common occupancy machinery and fixed-set estimates. The endpoint laws are proved in [Sections 3](#) to [5](#). [Sections 6](#) to [8](#) establish the global moving-lattice theorem and its uniform center. The perturbative and microscopic refinements appear in [Sections 9](#) and [10](#).

2. TRACES, OCCUPANCY, AND POISSON INTERPOLATION

2.1. External shattering. For a graph $G = (V, E)$ and $v \in V$, write

$$N(v) = \{u \in V : uv \in E\}, \quad N[v] = N(v) \cup \{v\}.$$

Definition 2.1. A set $S \subseteq V(G)$ is *externally shattered* if

$$\{N(v) \cap S : v \in V(G) \setminus S\} = 2^S.$$

External shattering implies both open- and closed-shattering. The converse can fail, but the internal vertices of a k -set can supply at most k traces. In the range considered here this correction is negligible in relative probability. We use *ordinary shattering* when a statement applies to either the open- or the closed-neighborhood convention.

Fix S with $|S| = k$. Call the symbol of probability q_n rare and the symbol of probability $1 - q_n$ common. A pattern with r common symbols has probability

$$\pi_r = q_n^k \rho_n^r,$$

and there are $\binom{k}{r}$ such patterns. With $m = n - k$ outside vertices, the minimum expected occupancy is $a_{n,k} = m q_n^k$.

If the number of outside vertices is Poisson with mean m , the occupancy counts of the 2^k patterns are independent. Hence the exact external coverage probability is

$$\prod_{r=0}^k (1 - e^{-a_{n,k} \rho_n^r})^{\binom{k}{r}} = e^{-\Phi_{n,k}}.$$

2.2. Relative de-Poissonization. The following elementary squeeze will be used repeatedly.

Lemma 2.2 (Monotone de-Poissonization). *For each n , let $b_n(r)$ be the probability of an increasing occupancy event with exactly r independent samples, and put*

$$Q_n(t) = \mathbb{E}[b_n(N_t)], \quad N_t \sim \text{Poisson}(t).$$

Suppose $m = m(n) \rightarrow \infty$ and $L = L(n) \gg \sqrt{m}$ satisfy

- (i) $Q_n(m \pm L) = (1 + o(1))Q_n(m)$;
- (ii) $\mathbb{P}(N_{m-L} > m) = o(Q_n(m))$.

Then

$$b_n(m) = (1 + o(1))Q_n(m).$$

The same assertion holds uniformly over any family of increasing events for which the two hypotheses hold uniformly.

Proof. Monotonicity gives

$$Q_n(m - L) \leq b_n(m) + \mathbb{P}(N_{m-L} > m)$$

and

$$Q_n(m + L) \geq b_n(m) \mathbb{P}(N_{m+L} \geq m).$$

Since $L/\sqrt{m} \rightarrow \infty$, the last probability is $1 - o(1)$. The assumptions squeeze $b_n(m)$ between $(1 - o(1))Q_n(m)$ and $(1 + o(1))Q_n(m)$. $\square$

2.3. A multi-partition interpolation. We need a comparison between simultaneous coverage of several cylinder partitions and the product of their marginal coverage probabilities. The following lemma controls all orders of the common Poisson process at once.

Lemma 2.3 (Poisson add-one identity). *Let Ω be finite. For $z \in \Omega$, let $Z_z(t)$ be independent Poisson variables with differentiable nonnegative means $\lambda_z(t)$. If f is bounded and $D_z f(x) = f(x + e_z) - f(x)$, then*

$$\frac{d}{dt} \mathbb{E} f(Z(t)) = \sum_{z \in \Omega} \lambda'_z(t) \mathbb{E} D_z f(Z(t)).$$

The identity holds one-sidedly at an endpoint when the means are affine and nonnegative.

Proof. For a single Poisson variable,

$$\frac{d}{d\lambda} \mathbb{E}g(N_\lambda) = \mathbb{E}[g(N_\lambda + 1) - g(N_\lambda)].$$

Apply this coordinate by coordinate and use the chain rule. $\square$

Lemma 2.4 (Multi-partition interpolation). *Fix $h \geq 1$. Let $\mathcal{C}_1, \dots, \mathcal{C}_h$ be finite measurable partitions of the same probability space, and let $\mathcal{R}_i \subseteq \mathcal{C}_i$ be the required cells of family i . A Poisson process of intensity m has cell means*

$$x_C = m\mu(C), \quad z_{CD} = m\mu(C \cap D) \quad (C \in \mathcal{C}_i, D \in \mathcal{C}_j).$$

Suppose every required cell has mean at least a_0 and $he^{-a_0} < 1$. If Q is the probability that every required cell of every family is hit, then

$$(7) \quad 0 \leq \log Q - \sum_{i=1}^h \sum_{C \in \mathcal{R}_i} \log(1 - e^{-x_C})$$

$$(8) \quad \leq \frac{2^h}{1 - he^{-a_0}} \sum_{1 \leq i < j \leq h} \sum_{C \in \mathcal{C}_i} \sum_{D \in \mathcal{C}_j} z_{CD} e^{-x_C - x_D + z_{CD}}.$$

The estimate is uniform in the required subcollections $\mathcal{R}_i$.

Proof. For every atom z of the common refinement, introduce one shared Poisson variable of mean $tm\mu(z)$ and one private variable for each family, each of mean $(1-t)m\mu(z)$. Family i sees the shared process and its own private process. Let $R(t)$ be the joint success probability. At $t = 0$ the families are independent, while $R(1) = Q$.

Fix an atom z . Before adding it, let a_i be the indicator that family i is successful, and let b_i be its value after adding z to that family. By Lemma 2.3, the contribution of z to $R'(t)/(m\mu(z))$ is

$$\mathbb{E} \left[\prod_i b_i + (h-1) \prod_i a_i - \sum_{i=1}^h b_i \prod_{j \neq i} a_j \right].$$

Because $a_i, b_i \in \{0, 1\}$ and $a_i \leq b_i$, the bracket is the indicator that the added atom repairs at least two failed families and leaves all families successful. In particular $R'(t) \geq 0$.

Decompose according to the exact repaired set $K \subseteq [h]$, $|K| \geq 2$. Let H_K be the increasing event that all non-target requirements are met, E_K the event that all target counts are zero, and P_K the event that all target counts are positive. Harris's inequality [5] and a union bound give

$$\mathbb{P}_t(H_K \cap E_K) \leq \mathbb{P}_t(H_K) \mathbb{P}_t(E_K), \quad R(t) \geq \mathbb{P}_t(H_K)(1 - he^{-a_0}).$$

For any pair $i, j \in K$ with target cells C, D ,

$$\mathbb{P}_t(E_K) \leq e^{-x_C - x_D + tz_{CD}} \leq e^{-x_C - x_D + z_{CD}}.$$

Summing over pairs in K , over the fewer than 2^h choices of K , and over the atom intensities inside $C \cap D$, we obtain

$$0 \leq \frac{R'(t)}{R(t)} \leq \frac{2^h}{1 - he^{-a_0}} \sum_{i < j} \sum_{C, D} z_{CD} e^{-x_C - x_D + z_{CD}}.$$

Integration from 0 to 1 proves (7). If a target cell is not required, its pivotal term is absent, proving uniformity in the required subcollections. $\square$

Lemma 2.5 (Uniform intensity stability). *Let $m \sim n$, let $k \rightarrow \infty$ with $k = O(\log n)$, and let $x_\sigma(m) = m\pi_\sigma$ be the means of the 2^k trace cells. Suppose*

$$a := \min_{\sigma} x_{\sigma}(m) \gg \log k, \quad \Phi(m) := \sum_{\sigma} F(x_{\sigma}(m)) = O(k^2).$$

Set $L = \sqrt{m}k^4$. Uniformly for $|t - m| \leq L$,

$$\sum_{\sigma} x_{\sigma}(t)e^{-x_{\sigma}(t)} = O(k^6),$$

$$\Phi(t) = \Phi(m) + o(1).$$

Proof. Split the cells into

$$\mathcal{A} = \{\sigma : x_{\sigma}(m) \leq 2k^4\}, \quad \mathcal{B} = \{\sigma : x_{\sigma}(m) > 2k^4\}.$$

For $\sigma \in \mathcal{A}$ and $|t - m| \leq L$,

$$|x_{\sigma}(t) - x_{\sigma}(m)| \leq \frac{L}{m} 2k^4 = O(k^8/\sqrt{n}) = o(1).$$

Thus $x_{\sigma}(t) \leq 3k^4$ and

$$\sum_{\sigma \in \mathcal{A}} x_{\sigma}(t)e^{-x_{\sigma}(t)} \leq 3k^4 e^{o(1)} \sum_{\sigma} e^{-x_{\sigma}(m)} = O(k^6),$$

because $\sum_{\sigma} e^{-x_{\sigma}(m)} \leq \Phi(m)$. For $\sigma \in \mathcal{B}$, we have $x_{\sigma}(t) \geq k^4$ for all large n , and hence

$$\sum_{\sigma \in \mathcal{B}} x_{\sigma}(t)e^{-x_{\sigma}(t)} \leq 2^k k^4 e^{-k^4} = o(1).$$

This proves the first estimate. Finally,

$$|\Phi'(t)| = \frac{1}{t} \sum_{\sigma} \frac{x_{\sigma}(t)}{e^{x_{\sigma}(t)} - 1} \leq \frac{2}{t} \sum_{\sigma} x_{\sigma}(t)e^{-x_{\sigma}(t)} = O(k^6/n).$$

Integration over an interval of length at most L proves $|\Phi(t) - \Phi(m)| = O(k^{10}/\sqrt{n}) = o(1)$. $\square$

2.4. Ordinary versus external shattering.

Lemma 2.6. *Let $k \rightarrow \infty$ with $k = O(\log n)$ and suppose $a_{n,k} \gg \log k$. If $\Phi_{n,k} = O(k^2)$, then, for a fixed k -set S ,*

$$\mathbb{P}(S \text{ is open-shattered}) = (1 + o(1))\mathbb{P}(S \text{ is externally shattered}),$$

and the same holds for closed shattering. The estimate is uniform in S . In particular,

$$\mathbb{P}(S \text{ is ordinary shattered but not externally shattered}) = o(e^{-\Phi_{n,k}}).$$

Proof. Put $m = n - k$ and $L = \sqrt{m}k^4$. Poissonize the outside vertices. Conditional on the graph induced by S , the vertices of S supply a deterministic set I of at most k trace patterns. At intensity t , let $x_{\sigma}(t) = t\pi_{\sigma}$ and let $Q_I(t)$ be the probability that every pattern outside I is hit. Then

$$\log Q_I(t) = -\Phi(t) + \sum_{\sigma \in I} F(x_{\sigma}(t)).$$

Uniformly for $|t - m| \leq L$, Lemma 2.5 gives $\Phi(t) = \Phi(m) + o(1)$, while the minimum cell mean is $(1 - o(1))a_{n,k}$. Consequently

$$0 \leq \sum_{\sigma \in I} F(x_{\sigma}(t)) \leq kF((1 - o(1))a_{n,k}) = o(1)$$

uniformly over all I with $|I| \leq k$. Hence

$$Q_I(m \pm L) = e^{-\Phi(m)+o(1)} = (1+o(1))Q_I(m)$$

uniformly in I . A Poisson Chernoff bound gives

$$\mathbb{P}(\text{Poisson}(m-L) > m) \leq \exp\{-\Omega(L^2/m)\} = e^{-\Omega(k^8)} = o(e^{-\Phi(m)}),$$

and therefore it is also $o(Q_I(m))$. Moreover $\mathbb{P}(\text{Poisson}(m+L) \geq m) = 1 - o(1)$. The uniform form of [Lemma 2.2](#) now shows that the fixed-sample outside coverage probability after the internal traces are supplied is $e^{-\Phi(m)+o(1)}$, uniformly in the internal graph. The same argument with $I = \emptyset$ gives the external probability. Averaging over the internal graph proves the relative comparison. Since external shattering implies ordinary shattering, subtraction gives the final assertion. $\square$

2.5. A general upper bound. The next lemma is used only when the one-set exponent is much larger than k^2 .

Lemma 2.7 (Empty-box upper bound). *Let $k = O(\log n)$, $m = n - k$, and let $x_\sigma = m\pi_\sigma$ be the means of the 2^k external trace cells. Assume $a = \min_\sigma x_\sigma \rightarrow \infty$, and put*

$$\Phi = \sum_\sigma -\log(1 - e^{-x_\sigma}), \quad H = \log \binom{n}{k}.$$

There is an absolute constant C such that, if

$$(9) \quad \Phi - H - Ck \log \left(e + \frac{\Phi}{k} \right) \longrightarrow +\infty,$$

then no k -set is open- or closed-shattered with high probability.

Proof. In the Poissonized model, the indicators that the cells are empty are independent Bernoulli variables with parameters $z_\sigma = e^{-x_\sigma}$. Let Z be their sum and put $w_\sigma = z_\sigma/(1 - z_\sigma)$. Then

$$\mathbb{P}(Z \leq k) = e^{-\Phi} \sum_{j=0}^k e_j((w_\sigma)_\sigma) \leq e^{-\Phi} \sum_{j=0}^k \frac{W^j}{j!}, \quad W = \sum_\sigma w_\sigma.$$

Since $a \rightarrow \infty$, $W \leq 2 \sum_\sigma z_\sigma \leq 2\Phi$ for all sufficiently large n . We claim that the last sum is at most

$$\exp \left\{ Ck \log \left(e + \frac{\Phi}{k} \right) \right\}$$

for an absolute C . Indeed, if $W \leq k$, then the sum is at most $e^W \leq e^k$. If $W > k$, then, using $j! \geq (j/e)^j$,

$$\sum_{j=0}^k \frac{W^j}{j!} \leq 1 + \sum_{j=1}^k \left(\frac{eW}{j} \right)^j \leq \exp \left\{ Ck \log \left(e + \frac{W}{k} \right) \right\}.$$

Since $W \leq 2\Phi$, the claim follows after changing C . The event $\{Z \leq k\}$ is increasing in the number of samples, so the fixed- m probability is at most $2+o(1)$ times the Poissonized probability. Ordinary shattering implies $Z \leq k$, since the k internal vertices supply at most k missing traces. A union bound over the $\binom{n}{k}$ candidate sets and (9) complete the proof. $\square$

2.6. Open and closed neighborhoods differ by at most one. We record the following elementary deterministic comparison.

Theorem 2.8. *For every finite graph G ,*

$$|D^\circ(G) - D^{\text{cl}}(G)| \leq 1.$$

Proof. The claim is immediate when the larger dimension is at most one, so assume $d \geq 2$. We prove $D^{\text{cl}}(G) \leq D^\circ(G) + 1$; the reverse inequality is identical. Suppose that a set $S \subseteq V(G)$ with $|S| = d$ is shattered by closed neighborhoods. Let

$$\mathcal{A} = \{N(v) \cap S : v \in V(G)\}$$

be the family of open-neighborhood traces on S . For every $v \notin S$ one has

$$N[v] \cap S = N(v) \cap S.$$

Thus the only closed traces that may differ from open traces are those produced by the d vertices of S themselves. Since the closed traces realize all 2^d subsets of S , the open trace family $\mathcal{A}$ contains at least $2^d - d$ distinct subsets of S .

If $\text{VCdim}(S, \mathcal{A}) \leq d - 2$, then the Sauer–Shelah lemma [8, 9] gives

$$|\mathcal{A}| \leq \sum_{i=0}^{d-2} \binom{d}{i} = 2^d - d - 1,$$

contradicting $|\mathcal{A}| \geq 2^d - d$. Hence $\text{VCdim}(S, \mathcal{A}) \geq d - 1$, and therefore $D^\circ(G) \geq d - 1$. This proves $D^{\text{cl}}(G) \leq D^\circ(G) + 1$. Interchanging open and closed neighborhoods gives the other inequality. $\square$

By Theorem 2.8, every bounded-window localization proved for one convention holds for the other. The sharper common critical laws require the probabilistic internal-witness estimates developed below.

2.7. Fixed-set occupancy in the uniform case. Assume first that $p = 1/2$. For each $v \in V \setminus S$, the trace $N(v) \cap S$ is uniformly distributed over the $B = 2^k$ subsets of S , independently for different v . Put

$$m = n - k, \quad u = \frac{m}{B}, \quad \lambda = Be^{-u}.$$

Lemma 2.9. *Suppose $m \sim n$, $B = \Theta(n/\log n)$, $u \rightarrow \infty$, $\lambda = O((\log n)^2)$, and $Be^{-2u} = o(1)$. Then*

$$\mathbb{P}(S \text{ is externally shattered}) = \exp(-\lambda + o(1)).$$

Proof. With a Poisson number of outside vertices of mean t , the B occupancy numbers are independent Poisson variables of mean t/B . Hence, at $t = m$,

$$Q(m) := \mathbb{P}(\text{all boxes are occupied}) = (1 - e^{-u})^B = \exp\{-Be^{-u} + O(Be^{-2u})\} = \exp(-\lambda + o(1)).$$

We verify the hypotheses of Lemma 2.2. Take

$$L = A\sqrt{m} \log n,$$

where A is a sufficiently large constant. Since in the present application $m \sim n$ and $B = \Theta(n/\log n)$, uniformly for $|t - m| \leq L$,

$$Be^{-t/B} = \lambda \exp\{-(t - m)/B\} = \lambda + o(1),$$

because $\lambda L/B = o(1)$, and

$$Be^{-2t/B} = O(\lambda^2/B) = o(1).$$

Thus $Q(t) = \exp\{-\lambda + o(1)\}$ uniformly on this interval. A Poisson Chernoff bound gives

$$\mathbb{P}(\text{Poi}(m - L) > m) \leq \exp\{-\Omega(A^2(\log n)^2)\} = o(e^{-\lambda})$$

after increasing A , since $\lambda = O((\log n)^2)$. Lemma 2.2 therefore gives the asserted fixed- m probability with relative error $1 + o(1)$. $\square$

Lemma 2.10. *Under the assumptions of Lemma 2.9, if additionally*

$$\lambda \frac{k}{2^k} = o(1),$$

then, for both open and closed neighborhoods,

$$\mathbb{P}(S \text{ is shattered}) = (1 + o(1))\mathbb{P}(S \text{ is externally shattered}) = \exp(-\lambda + o(1)),$$

and in particular

$$\mathbb{P}(S \text{ is shattered but not externally shattered}) = o(e^{-\lambda}).$$

Proof. Condition on the graph induced by S . For either neighborhood convention, the vertices of S then supply a fixed set I_S of at most k patterns, independently of the outside traces. In the Poissonized model the probability that every pattern outside I_S is hit is

$$(1 - e^{-u})^{B - |I_S|}.$$

Its ratio to full external coverage is

$$(1 - e^{-u})^{-|I_S|} = \exp\{O(ke^{-u})\} = \exp\left\{O\left(\lambda \frac{k}{B}\right)\right\} = 1 + o(1).$$

The estimates in the proof of Lemma 2.9, including their uniformity for $|t - m| \leq A\sqrt{m} \log n$, are unchanged after deleting at most k box requirements. Lemma 2.2 therefore de-Poissonizes this comparison uniformly in I_S . Averaging over the internal graph and using external shattering for the reverse inequality proves the claim. $\square$

2.8. A non-uniform occupancy estimate. For $p \neq 1/2$ the coupon probabilities are not uniform. Set

$$q = \min(p, 1 - p), \quad s = 1 - q, \quad \rho = \frac{s}{q}.$$

A pattern on S with r common symbols has probability $q^k \rho^r$ and there are $\binom{k}{r}$ such patterns. Let

$$a_k = (n - k)q^k$$

and

$$\Lambda_k = \sum_{r=0}^k \binom{k}{r} e^{-a_k \rho^r}.$$

The sharper exponent is

$$\Phi_k(p) = \sum_{r=0}^k \binom{k}{r} F(a_k \rho^r), \quad F(x) = -\log(1 - e^{-x}).$$

In the coarse localization regime, $a_k = \Theta(\log \log n)$ and $\Phi_k = (1 + o(1))\Lambda_k$ on the relevant layers.

The following estimate supplies the uniform tail control used in both de-Poissonization and the asymmetric proof. Write $L_1 = \log n$, $L_2 = \log \log n$, and $L_3 = \log \log \log n$.

Lemma 2.11. Fix $p \neq 1/2$. Suppose $c_-L_2 \leq a_k \leq c_+L_2$ for constants $0 < c_- < c_+ < \infty$. For every fixed $d > 0$ and nonnegative integer t , there is $C = C(p, c_-, c_+, d, t)$ such that

$$\sum_{r=0}^k \binom{k}{r} (a_k \rho^r)^t e^{-da_k \rho^r} \leq L_1^C.$$

If $a_k = cL_2 + O(1)$ and r is fixed, then

$$\binom{k}{r} e^{-a_k \rho^r} = L_1^{r - c\rho^r + o(1)}.$$

Moreover, for every fixed $M > 0$ there is a fixed $R = R(p, c_-, c_+, d, t, M)$ such that

$$\sum_{r=R}^k \binom{k}{r} (a_k \rho^r)^t e^{-da_k \rho^r} = O(L_1^{-M}).$$

Proof. The relation $a_k = (n - k)q^k = \Theta(L_2)$ implies $k = (L_1 - L_3 + O(1))/\log(1/q)$, so the fixed- r assertion follows from $\binom{k}{r} = L_1^{r + o(1)}$. For the uniform bound, choose a fixed R so large that

$$dc_- \rho^r \geq 3r + 3 \quad (r \geq R).$$

The sum over $r < R$ is a fixed sum of powers of L_1 and L_2 . For $r \geq R$, use $\binom{k}{r} \leq (ek/r)^r$. After increasing R once, the logarithm of the r -th summand is at most

$$r(L_2 + O(1)) + t\{O(L_3) + r \log \rho\} - dc_- L_2 \rho^r \leq -rL_2.$$

The remaining tail is summable and in fact $o(1)$. Increasing R until the last bound is at most $-(M + 2) \log L_1$ for every $r \geq R$ gives the stated $O(L_1^{-M})$ truncation. $\square$

Lemma 2.12. Suppose $k = O(\log n)$ and $a_k = \Theta(\log \log n)$. Then

$$\mathbb{P}(S \text{ is externally shattered}) = \exp\{-\Phi_k(p) + o(1)\}.$$

Proof. Let $m = n - k$. With a Poisson number of outside traces of mean t , each pattern in layer r is hit a Poisson number of times with mean $tq^k \rho^r$, independently over patterns. At $t = m$ the probability that all patterns are hit equals

$$\prod_{r=0}^k (1 - e^{-a_k \rho^r}) \binom{k}{r} = \exp\{-\Phi_k(p)\}.$$

We now verify relative de-Poissonization. Lemma 2.11, together with $F(x) \leq 2e^{-x}$ for $x \geq a_k \rightarrow \infty$, gives constants C_0, C_1 such that, uniformly for $|t - m| = o(m)$,

$$\Phi_k(t) := \sum_{r=0}^k \binom{k}{r} F(tq^k \rho^r) \leq L_1^{C_0},$$

and

$$\sum_{r=0}^k \binom{k}{r} (tq^k \rho^r) e^{-tq^k \rho^r/2} \leq L_1^{C_1}.$$

Choose $C > \max(C_0, C_1) + 3$ and $L = \sqrt{m} L_1^C = o(m)$. Differentiating the logarithm of the Poissonized probability gives

$$|\Phi'_k(t)| \leq \frac{2}{t} \sum_{r=0}^k \binom{k}{r} (tq^k \rho^r) e^{-tq^k \rho^r/2} \leq \frac{2L_1^{C_1}}{t}.$$

Hence $\Phi_k(m \pm L) = \Phi_k(m) + o(1)$. The Poisson tail in Lemma 2.2 is $\exp\{-\Omega(L_1^{2C})\} = o(e^{-\Phi_k(m)})$. That lemma now yields the displayed fixed- m asymptotic. $\square$

Lemma 2.13. *Assume $k = O(\log n)$, $a_k = \Theta(\log \log n)$, and $\Phi_k(p) = O((\log n)^2)$. Then, for open and closed neighborhoods,*

$$\mathbb{P}(S \text{ is shattered}) = (1 + o(1))\mathbb{P}(S \text{ is externally shattered}) = (1 + o(1))e^{-\Phi_k(p)}.$$

In particular,

$$\mathbb{P}(S \text{ is shattered but not externally shattered}) = o(e^{-\Phi_k(p)}).$$

Proof. Put $m = n - k$, let π_σ be the external probability of pattern σ , set $x_\sigma = m\pi_\sigma$, and write $F(x) = -\log(1 - e^{-x})$. Poissonize the outside traces. Conditional on the graph induced by S , let I be the set of patterns supplied by its internal vertices. Independence of internal and external edges gives the exact ratio

$$\frac{\mathbb{P}(\text{all patterns are supplied} \mid I)}{\mathbb{P}(\text{all patterns are supplied externally})} = \exp \left\{ \sum_{\sigma \in I} F(x_\sigma) \right\}.$$

Let τ_v be the trace supplied by $v \in S$. Its self-coordinate is forced, so its law is not exactly π_σ ; nevertheless, for both open and closed neighborhoods,

$$\mathbb{P}(\tau_v = \sigma) \leq \frac{\pi_\sigma}{q}.$$

Consequently, with $X = \sum_{v \in S} F(x_{\tau_v})$,

$$\mathbb{E}X \leq \frac{k}{qm} \sum_{\sigma} x_\sigma F(x_\sigma).$$

Also $0 \leq X \leq kF(a_k)$. Since $e^x - 1 \leq xe^b$ for $0 \leq x \leq b$,

$$1 \leq \frac{\mathbb{P}(\text{shattering after internal traces})}{e^{-\Phi_k(p)}} \leq 1 + e^{kF(a_k)} \frac{k}{qm} \sum_{\sigma} x_\sigma F(x_\sigma).$$

If $\theta = \log 2 / \log(1/q) < 1$, then $2^k = (m/a_k)^\theta$. Moreover $\sup_{x \geq a_k} xF(x) = O(1)$, $e^{kF(a_k)} = n^{o(1)}$, and hence the error above is at most

$$n^{o(1)} \frac{k2^k}{m} = n^{-(1-\theta)+o(1)} = o(1).$$

The same estimates hold uniformly at the two Poisson intensities used in the proof of Lemma 2.12. Applying Lemma 2.2 to the event averaged over the independent internal graph gives

$$\mathbb{P}(S \text{ is shattered}) = (1 + o(1))e^{-\Phi_k(p)}.$$

Lemma 2.12 gives the same relative asymptotic for external shattering. Since external shattering implies shattering, subtracting the two probabilities proves the final assertion. $\square$

3. THE SYMMETRIC CASE $p = 1/2$

In this section $G \sim G(n, 1/2)$ and

$$J_n = \left\lfloor \log_2 \frac{n}{\log n} \right\rfloor.$$

3.1. Two-point concentration.

Proposition 3.1. *For either open or closed neighborhoods,*

$$D_n \in \{J_n, J_n + 1\}$$

with high probability.

Proof. First fix any $S \subseteq V$ with $|S| = J_n$. Put $B = 2^{J_n}$ and $m = n - J_n$. Since $B \leq n/\log n$,

$$\frac{m}{B} \geq (1 - o(1)) \log n.$$

Thus

$$B \exp(-m/B) = o(1).$$

By Lemma 2.9, this fixed set is externally shattered with probability $1 - o(1)$. Hence $D_n \geq J_n$ with high probability.

For the upper bound, set $k = J_n + 2$ and fix $S \in \binom{V}{k}$. Then

$$2^k \in \left(\frac{2n}{\log n}, \frac{4n}{\log n} \right]$$

and therefore

$$\frac{n - k}{2^k} \leq \frac{1}{2} \log n (1 + o(1)).$$

Poissonize the outside sample with mean $m = n - k$, and let Z_S be the number of missing patterns there. Then Z_S is binomial and its mean satisfies

$$\mathbb{E}Z_S = (1 + o(1))2^k \exp\left(-\frac{n - k}{2^k}\right) \geq c \frac{\sqrt{n}}{\log n}$$

for some absolute $c > 0$. If S is shattered, all missing external patterns must be supplied internally, so $Z_S \leq k$. A Chernoff lower-tail estimate for the empty boxes gives

$$\mathbb{P}_{\text{Poi}(m)}(Z_S \leq k) \leq \exp\left(-c' \frac{\sqrt{n}}{\log n}\right).$$

The event $\{Z_S \leq k\}$ is increasing in the number of outside traces; hence its fixed- m probability is at most $2 + o(1)$ times the last display. This gives the same exponential bound for the probability that S is shattered. Since

$$\binom{n}{k} \leq \exp(O((\log n)^2)),$$

a union bound shows that no k -set is shattered with high probability. Hence $D_n \leq J_n + 1$ with high probability. $\square$

3.2. A cylinder decorrelation lemma. The following example shows why a direct Poissonized product comparison must be restricted to the critical range. Consider two sets differing in one vertex. If $S = R \cup \{x\}$ and $T = R \cup \{y\}$, $|R| = k - 1$, and the common outside sample is Poisson with mean m_0 , put $u_0 = m_0/B$ and $t = e^{-u_0}$. For each of the $B/2$ patterns on R , the four extensions form an independent 2×2 Poisson table. Inclusion-exclusion over its two empty rows and two empty columns gives

$$\mathbb{P}(S, T \text{ both covered by the common sample}) = (1 - 4t + 4t^{3/2} - t^2)^{B/2}.$$

Thus, when $Be^{-2u_0} = o(1)$,

$$\log \mathbb{P} = -2Be^{-u_0} + 2Be^{-3u_0/2} + o(1).$$

For example, let n_j be the least integer with $n_j/\log n_j \geq 2^j$. Then $J_{n_j} = j$ for all large j , $B = 2^{j+1} \sim 2n_j/\log n_j$, and $u_0 \sim \frac{1}{2} \log n_j$; hence $Be^{-3u_0/2} \asymp n_j^{1/4}/\log n_j \rightarrow \infty$. Thus the correction need not be negligible outside the range used below.

We control all orders at once by a Poisson interpolation.

Lemma 3.2. *Fix h . Let $S_1, \dots, S_h$ be distinct k -sets in a coordinate set U , give $\{0, 1\}^U$ uniform measure μ , and put $B = 2^k$. For each i , let $\mathcal{R}_i \subseteq 2^{S_i}$ be a collection of required patterns. A Poisson process of intensity m_0 is successful if it hits every cylinder*

$$C(i, \sigma) = \{z : z|_{S_i} = \sigma\}, \quad \sigma \in \mathcal{R}_i, \quad i \in [h].$$

Writing $u_0 = m_0/B$, if $he^{-u_0} < 1$, then

$$(10) \quad 0 \leq \log \mathbb{P}(\text{success}) - \sum_{i=1}^h |\mathcal{R}_i| \log(1 - e^{-u_0}) \leq \frac{2^h m_0 e^{-3u_0/2}}{1 - he^{-u_0}}.$$

The estimate is uniform in the required subcollections.

Proof. For every atom $z \in \{0, 1\}^U$, take one shared Poisson variable of mean $tm_0\mu(z)$ and, independently, one private variable for each family $i \in [h]$, each of mean $(1-t)m_0\mu(z)$. A shared point is counted in the cylinder system of every family, while a private point is counted only in its own family. Let $R(t)$ be the probability that every required cylinder is hit. At $t = 0$ the h systems are independent and their cells have independent $\text{Poi}(u_0)$ counts, whereas at $t = 1$ they are the projections of one common process. Hence

$$R(0) = \prod_{i=1}^h (1 - e^{-u_0})^{|\mathcal{R}_i|}, \quad R(1) = \mathbb{P}(\text{success}).$$

We bound the logarithmic derivative. Fix an atom z . Before adding it, let a_i be the indicator that family i is already successful, and let b_i be the corresponding indicator after adding z to that family. Lemma 2.3 shows that the contribution of z to $R'(t)/(m_0\mu(z))$ is the expectation of

$$\prod_i b_i + (h-1) \prod_i a_i - \sum_{i=1}^h b_i \prod_{j \neq i} a_j.$$

Because $a_i, b_i \in \{0, 1\}$ and $a_i \leq b_i$, this expression is exactly the indicator that adding z simultaneously repairs at least two families and that all families are successful afterwards. In particular $R'(t) \geq 0$.

Decompose this pivotal event according to its exact nonempty set $K \subseteq [h]$ of repaired families, where $|K| \geq 2$. For $i \in K$, the target cylinder $C_i = C(i, z|_{S_i})$ is required, empty, and every other required cylinder of family i is nonempty; every family outside K is already successful. Let H_K denote all these non-target nonemptiness requirements, let E_K be the event that the target counts are all zero, and let P_K be the event that they are all positive. Thus the pivotal event is $H_K \cap E_K$, while the full success event is $H_K \cap P_K$. Harris's inequality [5] for the underlying independent Poisson variables gives

$$\mathbb{P}_t(H_K \cap E_K) \leq \mathbb{P}_t(H_K) \mathbb{P}_t(E_K), \quad R(t) = \mathbb{P}_t(H_K \cap P_K) \geq \mathbb{P}_t(H_K) \mathbb{P}_t(P_K).$$

Every target count has marginal mean u_0 , so a union bound yields

$$\mathbb{P}_t(P_K) \geq 1 - |K|e^{-u_0} \geq 1 - he^{-u_0}.$$

The target cylinders all contain z , hence are compatible. For distinct i, j , since $S_i \neq S_j$ but $|S_i| = |S_j| = k$,

$$\mu(C_i \cap C_j) \leq \frac{1}{2B}, \quad \mu(C_i \cup C_j) \geq \frac{3}{2B}.$$

The private parts of the $|K|$ target counts have total mean $(1-t)|K|u_0$, and the shared part is zero precisely when the common process has no point in $\cup_{i \in K} C_i$. Consequently

$$\mathbb{P}_t(E_K) = \exp\{-(1-t)|K|u_0 - tm_0\mu(\cup_{i \in K} C_i)\} \leq e^{-3u_0/2}.$$

There are at most 2^h possible sets K . It follows that

$$0 \leq \frac{R'(t)}{R(t)} \leq \sum_z m_0 \mu(z) \frac{2^h e^{-3u_0/2}}{1 - h e^{-u_0}} = \frac{2^h m_0 e^{-3u_0/2}}{1 - h e^{-u_0}}.$$

Integrating from 0 to 1 proves (10). The argument only discards pivotal families whose target pattern is not required, so it is uniform in the subcollections $\mathcal{R}_i$. $\square$

Lemma 3.3. *Fix $h \geq 1$. Let $k = J_n + 1$, $B = 2^k$, $u = (n - k)/B$, and $\lambda = B e^{-u}$. Assume*

$$\lambda = O((\log n)^2), \quad B u e^{-3u/2} = o(1).$$

If $S_1, \dots, S_h$ are distinct k -subsets of V , then

$$\mathbb{P}(S_1, \dots, S_h \text{ are all externally shattered}) = \exp(-h\lambda + o(1)),$$

uniformly over the ordered h -tuples. Both assumptions hold automatically if $H_k - \lambda = O(1)$, and also if $H_k - \lambda \rightarrow +\infty$.

Proof. Let $U = S_1 \cup \dots \cup S_h$ and first use only the $m_0 = n - |U|$ vertices in $V \setminus U$. Since $|U| - k = O_h(k)$,

$$u_0 := m_0/B = u + O_h(k/B), \quad B e^{-u_0} = \lambda + o(1).$$

Poissonization and Lemma 3.2 give

$$\log Q(m_0) = hB \log(1 - e^{-u_0}) + O_h(B u_0 e^{-3u_0/2}) = -h\lambda + o(1).$$

This remains uniform for $|t - m_0| \leq L = A\sqrt{n} \log n$: indeed $L/B = o(1)$ and $\lambda L/B = o(1)$, while the interpolation error remains $o(1)$. The Poisson tail is $\exp\{-\Omega(A^2(\log n)^2)\} = o(e^{-h\lambda})$ for large fixed A . Lemma 2.2 therefore gives the same relative asymptotic with exactly m_0 common outside vertices.

It remains to restore the vertices in $U \setminus S_i$ as possible witnesses for S_i . Conditional on the graph induced by U , they supply at most $d = O_h(k)$ specified cylinder traces. Delete those requirements from the corresponding collections $\mathcal{R}_i$. Lemma 3.2 applies uniformly to both the full and deleted collections; their independent terms differ by

$$-d \log(1 - e^{-u_0}) = O(d e^{-u_0}) = O_h(k\lambda/B) = o(1),$$

and each interpolation error is $o(1)$. The same monotone squeeze de-Poissonizes these estimates uniformly. Since using only $V \setminus U$ is a subset of the true external-shattering event, the full and deleted requirements give matching lower and upper bounds.

Finally, if $H_k - \lambda = O(1)$ or tends to $+\infty$, then $\lambda \leq H_k + O(1) = O((\log n)^2)$ and

$$B u e^{-3u/2} = u \lambda^{3/2} B^{-1/2} = o(1),$$

because $B = \Theta(n/\log n)$. $\square$

3.3. Sharp threshold and Poisson law. We now prove the full Poisson critical window announced in Theorem 1.1. The key point is that, at the critical value $k = J_n + 1$, externally shattered k -sets occur as rare and asymptotically independent events. The independence is not literal, since two candidate sets may overlap, but Lemma 3.3 shows that all fixed factorial moments have the product form needed for the Poisson limit.

Let

$$k = J_n + 1, \quad B = 2^k, \quad u = \frac{n-k}{B}, \quad \lambda = Be^{-u},$$

and

$$H_k = \log \binom{n}{k}.$$

Let Y_k be the number of externally shattered k -sets.

By Lemma 2.9,

$$\mathbb{E}Y_k = \binom{n}{k} e^{-\lambda+o(1)} = \exp(H_k - \lambda + o(1)).$$

Proposition 3.4. *If $\lambda - H_k \rightarrow +\infty$, then $D_n = J_n$ with high probability. If $H_k - \lambda \rightarrow +\infty$, then $D_n = J_n + 1$ with high probability.*

Proof. If $\lambda - H_k \rightarrow +\infty$, then Lemma 2.10 gives

$$\mathbb{E}X_k \leq \exp(H_k - \lambda + o(1)) \rightarrow 0,$$

where X_k is the number of shattered k -sets, provided $\lambda \leq 10H_k$. The hypotheses of that lemma hold in this range because $H_k = \Theta((\log n)^2)$ and $B = \Theta(n/\log n)$. If instead $\lambda > 10H_k$, let Z_S be the number of externally empty boxes. In the Poissonized model $Z_S \sim \text{Bin}(B, e^{-u})$ with mean λ . Shattering implies $Z_S \leq k$, and monotonicity in the number of witnesses gives

$$\mathbb{P}_m(Z_S \leq k) \leq (2 + o(1))\mathbb{P}_{\text{Poi}(m)}(Z_S \leq k) \leq \exp(-\lambda/3)$$

for all sufficiently large n , since $k = o(\lambda)$. A union bound is then at most $\exp(H_k - \lambda/3) = o(1)$. Thus in both cases no k -set is shattered with high probability, and Proposition 3.1 gives $D_n = J_n$.

If $H_k - \lambda \rightarrow +\infty$, then $\mathbb{E}Y_k \rightarrow \infty$. By Lemma 3.3 with $h = 2$,

$$\mathbb{E}Y_k^2 = \mathbb{E}Y_k + (1 + o(1))(\mathbb{E}Y_k)^2.$$

The second-moment method yields $Y_k > 0$ with high probability, so $D_n \geq k = J_n + 1$. Proposition 3.1 gives equality. $\square$

Proposition 3.5. *If $H_k - \lambda \rightarrow \tau \in \mathbb{R}$, then*

$$Y_k \Rightarrow \text{Poisson}(e^\tau).$$

Consequently, for either open or closed neighborhoods,

$$\mathbb{P}(D_n = J_n + 1) \rightarrow 1 - e^{-e^\tau}.$$

Proof. For fixed $h \geq 1$,

$$\mathbb{E}[(Y_k)_h] = \sum_{(S_1, \dots, S_h)} \mathbb{P}(S_1, \dots, S_h \text{ externally shattered}),$$

where the sum is over ordered distinct k -sets. By Lemma 3.3,

$$\mathbb{E}[(Y_k)_h] = \left(\binom{n}{k} \right)^h \exp(-h\lambda + o(1)) = \exp(h(H_k - \lambda) + o(1)) \rightarrow e^{h\tau}.$$

These are the factorial moments of a Poisson random variable with mean e^τ , so $Y_k \Rightarrow \text{Poisson}(e^\tau)$.

By Lemma 2.10, the number of shattered but not externally shattered k -sets has expectation $o(1)$ in the critical window. Thus the count of open-shattered, and also closed-shattered, k -sets has the same limit. Since Proposition 3.1 gives $D_n \in \{J_n, J_n + 1\}$ with high probability, the displayed probability follows. $\square$

Theorem 1.1 follows from Propositions 3.1, 3.4, and 3.5.

Remark 3.6. The central critical scale $\lambda/H_k \rightarrow 1$ is equivalent to

$$\frac{n-k}{2^k} = \log n - 3 \log \log n + \log \log 2 + o(1).$$

More coarsely, $\lambda \asymp H_k$ is equivalent to the same display with $O(1)$ in place of $\log \log 2 + o(1)$. Indeed, $\lambda = (n-k)u^{-1}e^{-u}$ and $H_k = (1+o(1))(\log n)^2/\log 2$.

4. THE FIXED ASYMMETRIC CASE $p \neq 1/2$

We first prove the localization statement in Theorem 1.2; the critical-window refinement is given in Section 5. Throughout this section $p \neq 1/2$ is fixed, and

$$q = \min(p, 1-p), \quad \rho = \frac{1-q}{q}, \quad \alpha = \log(1/q).$$

We write

$$L_1 = \log n, \quad L_2 = \log \log n, \quad L_3 = \log \log \log n.$$

Thus

$$\log_{1/q} \log \log n = \frac{L_3}{\alpha}.$$

For a k -set S , call the symbol of probability q rare and the symbol of probability $1-q$ common. If a trace pattern on S has r common symbols, then its probability is

$$q^k \rho^r.$$

There are $\binom{k}{r}$ such patterns. For an integer sequence $k = k(n)$, define

$$a_k = (n-k)q^k,$$

and define the first-order missing-pattern sum

$$\Lambda_k = \sum_{r=0}^k \binom{k}{r} e^{-a_k \rho^r}.$$

The exact fixed-set exponent is

$$\Phi_k(p) = \sum_{r=0}^k \binom{k}{r} [-\log(1 - e^{-a_k \rho^r})],$$

but the coarser quantity Λ_k is sufficient for the $O_{\mathbb{P}}(1)$ localization.

4.1. The layer dichotomy. Recall

$$c_p = \max_{r \geq 3} \frac{r-2}{\rho^r}.$$

The maximum is finite and is attained, since $(r-2)/\rho^r \rightarrow 0$ as $r \rightarrow \infty$.

Lemma 4.1. *Fix $\varepsilon > 0$. Suppose*

$$k = \frac{L_1 - L_3 + O(1)}{\alpha}.$$

There exists $\delta = \delta(p, \varepsilon) > 0$ such that:

- (i) if $a_k \leq (c_p - \varepsilon)L_2$, then $\Lambda_k \geq L_1^{2+\delta}$;
- (ii) if $a_k \geq (c_p + \varepsilon)L_2$, then $\Lambda_k \leq L_1^{2-\delta}$.

Proof. For every fixed r ,

$$\binom{k}{r} e^{-a_k \rho^r} = L_1^{r - c\rho^r + o(1)}$$

whenever $a_k = cL_2 + O(1)$, since $k \sim L_1/\alpha$ and $L_2 = \log L_1$.

Assume first that $a_k \leq (c_p - \varepsilon)L_2$. Choose $r_* \geq 3$ such that

$$\frac{r_* - 2}{\rho^{r_*}} > c_p - \varepsilon/2.$$

Then the r_* -th layer alone gives

$$\Lambda_k \geq \binom{k}{r_*} e^{-a_k \rho^{r_*}} \geq L_1^{2+\delta}$$

for some $\delta = \delta(p, \varepsilon) > 0$.

Now assume that $a_k \geq (c_p + \varepsilon)L_2$. For every $r \geq 3$,

$$r - (c_p + \varepsilon)\rho^r \leq 2 - \varepsilon\rho^r.$$

For $r = 0, 1, 2$, the fixed-layer exponents are $-c$, $1 - c\rho$, and $2 - c\rho^2$, where $c \geq c_p + \varepsilon$; each is at most $2 - \delta_0$ for a fixed $\delta_0 > 0$. Choose a fixed R so large that

$$(c_p + \varepsilon)\rho^r \geq 3r \quad (r \geq R).$$

Using $\binom{k}{r} \leq (ek/r)^r$, exactly as in Lemma 2.11, shows that the sum over $r \geq R$ is $o(1)$. The remaining finite set of layers, together with the preceding exponent bounds, is at most $L_1^{2-\delta}$ after decreasing $\delta > 0$. $\square$

4.2. The upper bound.

Proposition 4.2. *Fix $\varepsilon > 0$. Suppose*

$$k = \frac{L_1 - L_3 + O(1)}{\alpha}$$

and

$$a_k = (n - k)q^k \leq (c_p - \varepsilon)L_2.$$

Then

$$\mathbb{P}(D_n^\circ(p) \geq k) \rightarrow 0, \quad \mathbb{P}(D_n^{\text{cl}}(p) \geq k) \rightarrow 0.$$

Proof. Fix $S \in \binom{V}{k}$ and let Z_S be the number of patterns missing from the traces of vertices in $V \setminus S$. In the Poissonized exposure, the missing indicators are independent and

$$\mathbb{E}Z_S = \Lambda_k.$$

By Lemma 4.1,

$$\Lambda_k \geq L_1^{2+\delta}.$$

If S is open- or closed-shattered, then all externally missing patterns must be supplied by the k vertices of S itself. Thus shattering implies

$$Z_S \leq k.$$

Since $k = O(L_1)$ and $k = o(\Lambda_k)$, Chernoff's lower-tail inequality in the Poissonized model gives

$$\mathbb{P}_{\text{Poi}(n-k)}(Z_S \leq k) \leq \exp(-c\Lambda_k)$$

for some $c > 0$. No relative de-Poissonization is needed for this upper bound. The event $\{Z_S \leq k\}$ is increasing in the number of outside traces, so if $N \sim \text{Poi}(n - k)$, then

$$\mathbb{P}_{n-k}(Z_S \leq k) \leq \frac{\mathbb{P}_{\text{Poi}(n-k)}(Z_S \leq k)}{\mathbb{P}(N \geq n - k)} \leq (2 + o(1))e^{-c\Lambda_k}.$$

Thus $\mathbb{P}(S \text{ is shattered}) \leq e^{-c'\Lambda_k}$ for some $c' > 0$.

Taking a union bound over all k -sets,

$$\mathbb{P}(\exists S, |S| = k, S \text{ shattered}) \leq \binom{n}{k} \exp(-c'\Lambda_k).$$

Here $\log \binom{n}{k} = O(L_1^2)$, while $\Lambda_k \geq L_1^{2+\delta}$, so the right-hand side tends to zero. The same argument applies to closed neighborhoods, because external open and closed traces agree and the internal vertices contribute at most k traces. $\square$

4.3. A packing lemma. The lower bound will use only a subfamily of candidate k -sets with controlled intersections. This avoids any need for a full non-uniform overlap-suppression theorem.

Lemma 4.3. *Let*

$$A \subseteq V, \quad |A| = N = \left\lfloor \frac{n}{L_1^3} \right\rfloor,$$

and let $J = L_1^{1-\eta}$ for a fixed $\eta > 0$. There is a family

$$\mathcal{F} \subseteq \binom{A}{k}$$

such that

$$|S \cap T| \leq J \quad (S \neq T \in \mathcal{F})$$

and

$$\log |\mathcal{F}| \geq J(L_1 - o(L_1)).$$

Proof. We use the greedy packing argument. Fix $S \in \binom{A}{k}$. The number of $T \in \binom{A}{k}$ with $|S \cap T| > J$ is at most

$$\sum_{j>J} \binom{k}{j} \binom{N-k}{k-j}.$$

Dividing by $\binom{N}{k}$, this is at most

$$\sum_{j>J} \left(\frac{Ck^2}{N} \right)^j$$

for an absolute constant C . Since $k = O(L_1)$ and $N = n/L_1^3$, we have

$$\frac{Ck^2}{N} = \exp(-L_1 + O(L_2)).$$

Thus the forbidden ball around S has size at most

$$\binom{N}{k} \exp(-J(L_1 - o(L_1))).$$

Greedy selection therefore produces a family of size at least

$$\exp(J(L_1 - o(L_1))).$$

$\square$

4.4. Small-overlap decorrelation. Let A be as in Lemma 4.3 and put

$$B_0 = V \setminus A, \quad M = |B_0| = (1 - o(1))n.$$

For $S \subseteq A$, we say that S is shattered by B_0 if

$$\{N(v) \cap S : v \in B_0\} = 2^S.$$

If this happens, then S is externally shattered in G .

For the bipartite exposure define

$$a'_k = Mq^k$$

and

$$\Phi'_k = \sum_{r=0}^k \binom{k}{r} [-\log(1 - e^{-a'_k \rho^r})].$$

The common multi-partition interpolation lemma, Lemma 2.4, will control the joint coverage probability.

Lemma 4.4. *Fix $\varepsilon > 0$ and assume*

$$k = \frac{L_1 - L_3 + O(1)}{\alpha}, \quad a'_k = Mq^k \geq (c_p + \varepsilon)L_2.$$

Then there exists $\delta = \delta(p, \varepsilon) > 0$ such that

$$\Phi'_k \leq L_1^{2-\delta}.$$

Moreover, for every fixed $\eta > 0$, if

$$|S \cap T| \leq J = L_1^{1-\eta}, \quad S, T \in \binom{A}{k},$$

then

$$\mathbb{P}(S, T \text{ are both shattered by } B_0) = \exp(-2\Phi'_k + o(1)),$$

uniformly over such pairs.

Proof. Because $k = (L_1 - L_3 + O(1))/\alpha$, we also have $q^k = \Theta(L_2/n)$ and hence

$$a'_k - a_k = (M - n + k)q^k = o(1).$$

Thus the layer dichotomy applies with $\varepsilon/2$, and $a'_k = \Theta(L_2)$. Uniformly for $x \geq a'_k \rightarrow \infty$, $F(x) = (1 + o(1))e^{-x}$. Lemmas 4.1 and 2.11 therefore give, after decreasing $\delta > 0$,

$$\Phi'_k \leq L_1^{2-\delta}.$$

Poissonize the number of vertices in B_0 with mean M . For cylinders C on S and D on T , put

$$x_C = M\mu(C), \quad y_D = M\mu(D), \quad z_{CD} = M\mu(C \cap D).$$

If C, D disagree on $S \cap T$, then $z_{CD} = 0$. Otherwise, if h is the probability of their common pattern on $S \cap T$, product structure gives

$$z_{CD} = \frac{x_C y_D}{Mh} \leq \frac{x_C y_D}{Mq^j}, \quad j = |S \cap T|.$$

Conditional on one trace pattern, the $k - j$ remaining coordinates are still independent, so also

$$\frac{z_{CD}}{x_C} \leq s^{k-j}, \quad \frac{z_{CD}}{y_D} \leq s^{k-j}, \quad s = 1 - q.$$

Since $j \leq L_1^{1-\eta} = o(L_1)$,

$$s^{k-j} = n^{-\gamma+o(1)}, \quad \gamma = \frac{\log(1/s)}{\log(1/q)} > 0.$$

In particular, uniformly for all compatible pairs, $z_{CD} \leq \frac{1}{2} \min(x_C, y_D)$ for large n .

Put

$$P_* := \sum_C x_C e^{-x_C/2} = \sum_{r=0}^k \binom{k}{r} (a'_k \rho^r) e^{-a'_k \rho^r/2}.$$

Lemma 2.11 gives $P_* \leq L_1^{C_0}$ for a fixed C_0 ; the same bound holds for the D -sum. Therefore

$$\begin{aligned} \sum_{C,D} z_{CD} e^{-x_C - y_D + z_{CD}} &\leq \frac{1}{Mq^j} \left(\sum_C x_C e^{-x_C/2} \right) \left(\sum_D y_D e^{-y_D/2} \right) \\ &\leq \frac{L_1^{2C_0}}{Mq^j} = n^{-1+o(1)}, \end{aligned}$$

because $Mq^j = \exp\{L_1 - \alpha j + o(L_1)\} = n^{1-o(1)}$. The minimum cell mean is $a'_k \rightarrow \infty$, so Lemma 2.4 with $h = 2$ yields

$$\mathbb{P}_{\text{Poi}(M)}(S, T \text{ both shattered by } B_0) = \exp(-2\Phi'_k + o(1))$$

uniformly over the stated pairs. The interpolation lemma controls all higher correlations at once; no truncation at pair clusters is being used.

Finally choose $L = \sqrt{M} L_1^C$ with C sufficiently large. All preceding estimates hold uniformly after replacing M by $M \pm L$, and the derivative estimate from Lemma 2.12 gives $\Phi'_k(M \pm L) = \Phi'_k(M) + o(1)$. A Poisson Chernoff tail is then $o(\exp\{-2\Phi'_k\})$. Applying Lemma 2.2 to the joint increasing event proves the fixed- M assertion. $\square$

4.5. The lower bound.

Proposition 4.5. *Fix $\varepsilon > 0$. Suppose*

$$k = \frac{L_1 - L_3 + O(1)}{\alpha}$$

and

$$Mq^k \geq (c_p + \varepsilon)L_2,$$

where $M = |V \setminus A|$ and $|A| = \lfloor n/L_1^3 \rfloor$. Then

$$\mathbb{P}(D_n^\circ(p) \geq k) \rightarrow 1, \quad \mathbb{P}(D_n^{\text{cl}}(p) \geq k) \rightarrow 1.$$

Proof. Let $\delta > 0$ be supplied by Lemma 4.4. Choose $\eta > 0$ with $\eta < \delta$, put

$$J = L_1^{1-\eta},$$

and take a packing $\mathcal{F} \subseteq \binom{A}{k}$ as in Lemma 4.3. Let Y be the number of sets $S \in \mathcal{F}$ shattered by B_0 .

For a fixed $S \in \mathcal{F}$,

$$\mathbb{P}(S \text{ is shattered by } B_0) = \exp(-\Phi'_k + o(1)),$$

and Lemma 4.4 gives $\Phi'_k \leq L_1^{2-\delta}$. Since

$$\log |\mathcal{F}| \geq J(L_1 - o(L_1)) = L_1^{2-\eta+o(1)},$$

we get

$$\log \mathbb{E}Y \geq L_1^{2-\eta+o(1)} - L_1^{2-\delta} \rightarrow +\infty.$$

Thus $\mathbb{E}Y \rightarrow \infty$.

For distinct $S, T \in \mathcal{F}$, Lemma 4.4 gives

$$\mathbb{P}(S, T \text{ both shattered by } B_0) = \exp(-2\Phi'_k + o(1)).$$

Therefore

$$\mathbb{E}Y^2 = \mathbb{E}Y + (1 + o(1))|\mathcal{F}|^2 e^{-2\Phi'_k + o(1)} = \mathbb{E}Y + (1 + o(1))(\mathbb{E}Y)^2.$$

Since $\mathbb{E}Y \rightarrow \infty$,

$$\frac{\mathbb{E}Y^2}{(\mathbb{E}Y)^2} \rightarrow 1.$$

By the second-moment method, $Y > 0$ with high probability. Hence some k -set is externally shattered, which implies both the open and closed neighborhood VC-dimensions are at least k . $\square$

4.6. Proof of Theorem 1.2. Set

$$\kappa_n = \log_{1/q} n - \log_{1/q} \log \log n - \log_{1/q} c_p.$$

Equivalently,

$$\kappa_n = \frac{L_1 - L_3 - \log c_p}{\alpha}, \quad nq^{\kappa_n} = c_p L_2.$$

Choose $C > 0$ large. For

$$k_+ = \lceil \kappa_n + C \rceil$$

we have

$$(n - k_+)q^{k_+} = c_p L_2 q^{C+O(1)}(1 + o(1)).$$

For C large enough this is at most $(c_p - \varepsilon)L_2$, so Proposition 4.2 gives $D_n(p) < k_+$ with high probability.

Similarly, for

$$k_- = \lfloor \kappa_n - C \rfloor,$$

we have

$$nq^{k_-} = c_p L_2 q^{-C+O(1)}(1 + o(1)).$$

For C large enough this is at least $(c_p + \varepsilon)L_2$. Since $M = (1 - o(1))n$, Proposition 4.5 gives $D_n(p) \geq k_-$ with high probability.

Thus

$$D_n(p) = \kappa_n + O_{\mathbb{P}}(1),$$

which is precisely the assertion of Theorem 1.2. The argument applies to both open and closed neighborhoods.

4.7. Structure of the dominant rare layers. In the asymmetric theorem, the second-order constant is

$$c_p = \max_{r \geq 3} \frac{r-2}{\rho^r}, \quad \rho = \frac{1-q}{q} > 1.$$

The maximizing values of r are exactly the rare layers that determine the leading obstruction to shattering.

Theorem 4.6. *For $\rho > 1$, let $f_\rho(r) = (r-2)/\rho^r$ for integers $r \geq 3$, and put $R(\rho) = \arg \max_{r \geq 3} f_\rho(r)$. Then $R(\rho)$ consists of one integer or two adjacent integers. If $\rho > 2$, then $R(\rho) = \{3\}$; if $\rho = 2$, then $R(\rho) = \{3, 4\}$. For $1 < \rho < 2$, an integer $r \geq 4$ is a maximizer precisely when*

$$\frac{r-1}{r-2} \leq \rho \leq \frac{r-2}{r-3}.$$

Moreover,

$$R(\rho) \subseteq \left\{ \left\lfloor 2 + \frac{1}{\log \rho} \right\rfloor, \left\lceil 2 + \frac{1}{\log \rho} \right\rceil \right\}.$$

For $\rho > 1$, define

$$f_\rho(r) = \frac{r-2}{\rho^r}, \quad r = 3, 4, 5, \dots,$$

and

$$R(\rho) = \arg \max_{r \geq 3} f_\rho(r).$$

Lemma 4.7. *The sequence $f_\rho(r)$ is unimodal. Moreover,*

$$\frac{f_\rho(r+1)}{f_\rho(r)} = \frac{r-1}{r-2} \cdot \frac{1}{\rho}.$$

Consequently $f_\rho(r+1) \geq f_\rho(r)$ if and only if

$$\rho \leq \frac{r-1}{r-2}.$$

Proof. The displayed ratio follows immediately from the definition:

$$\frac{f_\rho(r+1)}{f_\rho(r)} = \frac{(r-1)\rho^{-r-1}}{(r-2)\rho^{-r}} = \frac{r-1}{r-2} \cdot \frac{1}{\rho}.$$

The factor $(r-1)/(r-2)$ decreases strictly to 1 as r increases. Since $1/\rho < 1$, the ratio crosses 1 at most once. Thus the sequence first increases and then decreases, with the possibility of a single adjacent tie. This is unimodality. $\square$

Proof of Theorem 4.6. By Lemma 4.7, the set of global maximizers contains either one integer or two adjacent integers.

If $\rho > 2$, then

$$\frac{f_\rho(4)}{f_\rho(3)} = \frac{2}{\rho} < 1.$$

By unimodality the sequence is strictly decreasing after $r = 3$, so $R(\rho) = \{3\}$. If $\rho = 2$, then $f_\rho(3) = f_\rho(4)$ and, again by unimodality, $R(\rho) = \{3, 4\}$.

Now take $1 < \rho < 2$. For $r \geq 4$, the integer r is a maximizer if and only if it is at least as large as its two neighbors:

$$f_\rho(r) \geq f_\rho(r-1), \quad f_\rho(r) \geq f_\rho(r+1).$$

Using the ratio formula, these two inequalities are equivalent to

$$\frac{f_\rho(r)}{f_\rho(r-1)} = \frac{r-2}{r-3} \cdot \frac{1}{\rho} \geq 1$$

and

$$\frac{f_\rho(r+1)}{f_\rho(r)} = \frac{r-1}{r-2} \cdot \frac{1}{\rho} \leq 1.$$

Equivalently,

$$\frac{r-1}{r-2} \leq \rho \leq \frac{r-2}{r-3}.$$

At the left endpoint, $f_\rho(r+1) = f_\rho(r)$, giving the adjacent tie $\{r, r+1\}$. At the right endpoint, $f_\rho(r) = f_\rho(r-1)$, giving the adjacent tie $\{r-1, r\}$.

Finally, the continuous relaxation

$$f_\rho(x) = (x-2)\rho^{-x}, \quad x > 2,$$

has logarithmic derivative

$$\frac{d}{dx} \log f_\rho(x) = \frac{1}{x-2} - \log \rho.$$

The unique critical point is therefore

$$x_\rho = 2 + \frac{1}{\log \rho},$$

which is the continuous maximum. Unimodality therefore places every discrete maximizer among the two integers bracketing x_ρ . $\square$

The dominant layer set $R(\rho)$ changes only when

$$\rho = \frac{r-1}{r-2}$$

for some $r \geq 3$, because these are precisely the equality cases in the ratio test. In particular, $R(2) = \{3, 4\}$ and $c_p = 1/8$, whereas $R(\rho) = \{3\}$ for $\rho > 2$.

5. THE ASYMMETRIC POISSON CRITICAL WINDOW

We now sharpen Theorem 1.2 by proving a Poisson law at the critical lattice points. Fix $p \neq 1/2$ throughout this section and retain

$$q = \min(p, 1-p), \quad s = 1-q, \quad \rho = s/q, \quad \alpha = \log(1/q).$$

Write $F(x) = -\log(1 - e^{-x})$. For $k = k(n)$ put

$$a = (n-k)q^k, \quad H_k = \log \binom{n}{k}, \quad \Phi_k = \sum_{r=0}^k \binom{k}{r} F(a\rho^r),$$

and let Y_k be the number of externally shattered k -sets. We now prove Theorem 1.3.

5.1. Scaling forced by the critical equation.

Lemma 5.1. *Under the hypotheses of Theorem 1.3,*

$$(11) \quad a = c_p \log k + O(1) = c_p \log \log n + O(1).$$

Proof. The assumption $a = \Theta(\log \log n)$ gives

$$k = \frac{\log n - \log \log \log n + O(1)}{\alpha}, \quad H_k = \alpha k^2 + o(k^2).$$

Since $a \rightarrow \infty$, $F(x) = (1 + o(1))e^{-x}$ uniformly for $x \geq a$. Put $c_n = a/\log k$. The sequence (c_n) stays in a compact subset of $(0, \infty)$. Along any subsequence on which $c_n \rightarrow c$, every fixed layer satisfies

$$\binom{k}{r} F(a\rho^r) = k^{r-c\rho^r+o(1)},$$

and Lemma 2.11 supplies a uniform finite truncation. If $c < c_p$, a maximizing layer has exponent larger than 2; if $c > c_p$, every layer has exponent bounded below 2, uniformly after truncation. Since $\Phi_k = H_k + O(1) = \Theta(k^2)$, neither alternative is possible. Every subsequential limit is therefore c_p , and $a = c_p \log k + o(\log k)$.

Write $a = c_p \log k + b_n$. For every maximizing layer $r \in R(\rho)$,

$$\binom{k}{r} F(a\rho^r) = \frac{k^2}{r!} e^{-b_n \rho^r} (1 + o(1)),$$

whereas every fixed nonmaximizing layer is $k^{2-\delta_r+o(1)}$ and the large layers are controlled by the same exponential tail. If $b_n \rightarrow +\infty$, then $\Phi_k = o(k^2)$; if $b_n \rightarrow -\infty$, then $\Phi_k/k^2 \rightarrow \infty$. Thus $b_n = O(1)$, proving (11). $\square$

5.2. All overlap scales. Let S, T be distinct k -sets, put

$$j = |S \cap T|, \quad d = k - j \in \{1, \dots, k\},$$

and use a Poisson intensity $m = n - O(k)$, with $a_m = mq^k$. If compatible cells have t common symbols on $S \cap T$, b on $S \setminus T$, and c on $T \setminus S$, their means are

$$x = a_m \rho^{t+b}, \quad y = a_m \rho^{t+c}, \quad z = a_m q^d \rho^{t+b+c}.$$

The pair correction from Lemma 2.4 is therefore

$$\Delta_j = \sum_{t=0}^j \sum_{b,c=0}^d \binom{j}{t} \binom{d}{b} \binom{d}{c} a_m q^d \rho^{t+b+c} e^{-a_m G(t,b,c,d)},$$

where

$$G(t, b, c, d) = \rho^{t+b} + \rho^{t+c} - q^d \rho^{t+b+c}.$$

Lemma 5.2. *There is $\delta_p > 0$ such that, for every admissible t, b, c, d ,*

$$t - c_p G(t, b, c, d) \leq 2 - \delta_p.$$

Proof. Assume $b \geq c$. If $b = c = 0$, then $\rho^b + \rho^c - q^d \rho^{b+c} = 2 - q^d \geq 2 - q$. If $b \geq 1$, then $d \geq b$ and

$$q^d \rho^{b+c} \leq q^b \rho^{b+c} = s^b \rho^c,$$

so

$$\rho^b + \rho^c - q^d \rho^{b+c} \geq \rho^b + \rho^c (1 - s^b) \geq \rho + q \geq 2 - q.$$

Thus $G(t, b, c, d) \geq (2 - q)\rho^t$. For $t = 0, 1$ the desired left side is at most 1. For $t = 2$ it is at most $2 - c_p(2 - q)\rho^2 < 2$. For $t \geq 3$, the definition of c_p gives $c_p \rho^t \geq t - 2$, whence

$$t - c_p G(t, b, c, d) \leq t - (2 - q)(t - 2) = 2 - (1 - q)(t - 2) \leq 2 - (1 - q).$$

Taking the minimum of these fixed gaps proves the lemma. $\square$

Lemma 5.3. *Uniformly over all two distinct k -sets,*

$$(12) \quad \Delta_0 = o(1), \quad \sup_{1 \leq j \leq k-1} \frac{\Delta_j}{j \log n} = o(1).$$

Proof. First suppose $d \geq C \log k$. For compatible cells let h be the probability of their common pattern. Product structure gives

$$z_{CD} = \frac{x_C y_D}{mh} \leq \frac{x_C y_D}{mq^j}, \quad \frac{z_{CD}}{x_C}, \frac{z_{CD}}{y_D} \leq s^d.$$

For large C we have $s^d \leq 1/2$. With

$$P_* = \sum_C x_C e^{-x_C/2} \leq k^{A_p}$$

by the exponential layer-tail estimate,

$$\Delta_j \leq \frac{P_*^2}{mq^j} = \frac{q^d}{a_m} P_*^2 = o(1)$$

after choosing $C > 3A_p/\alpha$. This includes $j = 0$.

Now suppose $1 \leq d < C \log k$. We first justify a uniform finite truncation of the three layer indices. Given $M > 0$, choose a fixed integer $T = T(M, p)$, independent of d and n .

For the range $t > T$, use the stronger bound $G(t, b, c, d) \geq (2 - q)\rho^t$ and sum the b, c factors exactly:

$$\sum_{b,c=0}^d \binom{d}{b} \binom{d}{c} q^d \rho^{b+c} = q^d (1 + \rho)^{2d} = q^{-d}.$$

Since $d < C \log k$, the contribution with $t > T$ is at most

$$a_m q^{-d} \sum_{t>T} \binom{j}{t} \rho^t e^{-(2-q)a_m \rho^t}.$$

For all large n we have $a_m \geq (c_p/2) \log k$, while $q^{-d} \leq k^{C\alpha+o(1)}$. If

$$v_t = \binom{j}{t} \rho^t e^{-(2-q)a_m \rho^t},$$

then

$$\frac{v_{t+1}}{v_t} \leq k \rho k^{-\frac{1}{2}(2-q)c_p(\rho-1)\rho^t}.$$

Choose T so large that $\frac{1}{2}(2-q)c_p \rho^{T+1} > M + C\alpha + T + 4$ and the displayed ratio is at most k^{-2} for all $t \geq T + 1$. The first term of the tail is then $k^{-M-C\alpha-3}$, and the geometric ratio bound shows that the complete $t > T$ contribution is $O(k^{-M})$.

It remains to truncate b and c when $t \leq T$. From

$$G(t, b, c, d) \geq q(\rho^{t+b} + \rho^{t+c})$$

we may sum over c after discarding its exponential factor:

$$q^d \sum_{c=0}^d \binom{d}{c} \rho^c = q^d (1 + \rho)^d = 1.$$

Thus, apart from the fixed factor $a_m \binom{j}{t} \rho^t = k^{t+o(1)}$, the $b > T$ contribution is bounded by

$$\sum_{b>T} \binom{d}{b} \rho^b e^{-a_m q \rho^{t+b}} \leq \sum_{b>T} (C \log k)^b \rho^b k^{-\frac{1}{2}c_p q \rho^{t+b}}.$$

The ratio of consecutive summands is at most

$$C \rho \log k k^{-\frac{1}{2}c_p q(\rho-1)\rho^{t+b}}.$$

After increasing T , the first summand is at most k^{-M-T-3} and the ratio is at most k^{-2} , uniformly for every $t \leq T$. Hence the $b > T$ tail is $O(k^{-M})$ after restoring the fixed factor. The same argument applies to $c > T$. This proves a truncation uniform in $d < C \log k$. For each of the finitely many remaining triples,

$$\binom{j}{t} \binom{d}{b} \binom{d}{c} a_m q^d \rho^{t+b+c} e^{-a_m G(t,b,c,d)} \leq k^{t-c_p G(t,b,c,d)+o(1)} \leq k^{2-\delta_p+o(1)}$$

by Lemma 5.2. Hence $\Delta_j \leq k^{2-\delta}$ for some $\delta > 0$. In this range $j = k - O(\log k) \sim k$ and $\log n \sim \alpha k$, so this is $o(j \log n)$. Together with the first range this proves (12). $\square$

5.3. Joint probabilities and factorial moments.

Lemma 5.4. *Fix h and let $S_1, \dots, S_h$ be distinct k -sets. If $j_{i\ell} = |S_i \cap S_\ell|$, then*

$$\mathbb{P}(S_1, \dots, S_h \text{ are all externally shattered}) \leq \exp \left\{ -h\Phi_k + o(1) + C_h \sum_{i < \ell} \Delta_{j_{i\ell}} \right\}.$$

If the sets are pairwise disjoint, equality holds with exponent $-h\Phi_k + o(1)$. All errors are uniform.

Proof. The case $h = 1$ is Lemma 2.12, so assume $h \geq 2$. Let $U = \bigcup_i S_i$ and first use only the $m = n - |U|$ vertices outside U . Changing the intensity from $n - k$ by $O_h(k)$ changes each marginal logarithm by $o(1)$, because

$$\left| \frac{d}{dm} \Phi_k(m) \right| \leq \frac{2}{m} \sum_r \binom{k}{r} (mq^k \rho^r) e^{-mq^k \rho^r / 2} \leq \frac{k^{A_p}}{m}.$$

Lemma 2.4 gives the Poissonized joint upper bound.

We must restore the special witnesses in $U \setminus S_i$. Condition on all edges induced by U , and let I_i be the cells already supplied for family i . Deleting these requirements multiplies the independent baseline by e^X , where

$$X = \sum_i \sum_{\sigma \in I_i} F(x_\sigma).$$

There are $O_h(k)$ such traces and each has marginal product law π_σ . Therefore

$$\mathbb{E}X \leq \frac{C_h k}{m} \sum_\sigma x_\sigma F(x_\sigma) \leq n^{-(1-\theta)+o(1)}, \quad \theta = \frac{\log 2}{\log(1/q)} < 1.$$

Also $X \leq C_h k F(a_m)$ and $e^{C_h k F(a_m)} = n^{o(1)}$, so

$$\mathbb{E}e^X = 1 + o(1).$$

The interpolation estimate is uniform in deleted requirements, proving the Poissonized upper bound after averaging the special witnesses.

It remains to pass to exactly m common witnesses. Take $L = \sqrt{n} k^D$ with a fixed $D > 1$. The marginal derivative displayed above already gives additive $o(1)$ stability between $m - L$ and $m + L$. For the pair terms, let μ denote the Bernoulli product measure on the relevant union of coordinates and write

$$\iota_{CD} = \mu(C \cap D), \quad v_{CD} = \mu(C \cup D).$$

At intensity m' ,

$$\Delta_j(m') = \sum_{C,D} m' \iota_{CD} e^{-m' v_{CD}},$$

and hence

$$\left| \frac{d}{dm'} \Delta_j(m') \right| \leq \frac{1}{m'} \sum_{C,D} z_{CD} (1 + m' v_{CD}) e^{-m' v_{CD}}.$$

The two-range argument of Lemma 5.3, with the harmless extra factor $1 + m' v_{CD}$, gives a fixed A'_p such that

$$\sup_{0 \leq j \leq k-1} \sup_{|m' - m| \leq L} \left| \frac{d}{dm'} \Delta_j(m') \right| \leq \frac{k^{A'_p}}{m'}.$$

Indeed, for $d \geq C \log k$ one uses the weighted layer sums $\sum_C x_C(1 + x_C)e^{-x_C/2} \leq k^{O_p(1)}$; for $d < C \log k$ the exponential tail absorbs the extra polynomial factor. Consequently

$$\sup_j |\Delta_j(m \pm L) - \Delta_j(m)| \leq \frac{Lk^{A'_p}}{m - L} = o(1).$$

The minimum cell mean throughout this interval is $(m - L)q^k \sim c_p \log k$, so the interpolation denominator also tends to one.

We use monotonicity in one-sided form. Let $g(r)$ be the joint success probability with r common witnesses, after averaging the special witnesses, and put $Q(t) = \mathbb{E}[g(\text{Poi}(t))]$. If $N_+ \sim \text{Poi}(m + L)$, then

$$g(m) \leq \frac{Q(m + L)}{\mathbb{P}(N_+ \geq m)} = (1 + o(1))Q(m + L),$$

which transfers the required upper bound. For pairwise disjoint sets let g_0 and Q_0 denote success using only $V \setminus U$. With $N_- \sim \text{Poi}(m - L)$,

$$g_0(m) \geq Q_0(m - L) - \mathbb{P}(N_- > m).$$

Here Lemma 5.3 gives $\Delta_0 = o(1)$, so $Q_0(m - L) = e^{-h\Phi_k + o(1)}$. Both Poisson tails are $e^{-\Omega(k^{2D})} = o(e^{-h\Phi_k})$, because $\Phi_k = \Theta(k^2)$. Since the true event contains success using only $V \setminus U$, this proves the matching lower bound and completes the fixed-sample estimate. $\square$

Proposition 5.5. *For every fixed $h \geq 1$,*

$$\mathbb{E}[(Y_k)_h] \longrightarrow e^{h\tau}.$$

Proof. The number of ordered pairwise disjoint h -tuples is $(1 + o(1))\binom{n}{k}^h$. Lemma 5.4 shows that their contribution tends to $e^{h\tau}$.

For an ordered tuple set

$$U_i = S_1 \cup \dots \cup S_i, \quad r_i = |S_i \cap U_{i-1}| \ (i \geq 2), \quad R = \sum_{i=2}^h r_i.$$

For prescribed $r_2, \dots, r_h$, the proportion of such tuples among all ordered tuples is at most

$$(13) \quad \prod_{i=2}^h \left(\frac{C_h k^2}{n} \right)^{r_i}.$$

Moreover

$$\sum_{i < \ell} |S_i \cap S_\ell| \leq hR.$$

By Lemma 5.3, for some $\epsilon_n \rightarrow 0$, $\Delta_j \leq \epsilon_n j \log n$ when $j \geq 1$, while $\Delta_0 = o(1)$. Thus the joint probability of a tuple with $R \geq 1$ is at most

$$e^{-h\Phi_k + o(1)} e^{o(R \log n)}.$$

Summing this with (13), the overlapping contribution relative to $\binom{n}{k}^h e^{-h\Phi_k}$ is bounded by a convergent geometric sum with ratio $n^{-1+o(1)}$, and hence is $o(1)$. The disjoint contribution proves the proposition. $\square$

Proof of Theorem 1.3. Proposition 5.5 gives all factorial moments of $\text{Poisson}(e^\tau)$, proving the asserted convergence. The final assertion of Lemma 2.13 gives, for every fixed k -set and hence uniformly over its choice by symmetry,

$$\mathbb{P}(S \text{ is shattered but not externally shattered}) = o(e^{-\Phi_k}).$$

Since $\binom{n}{k}e^{-\Phi_k} \rightarrow e^\tau$, the external and ordinary counts agree with probability $1 - o(1)$ at the critical level.

By Lemma 5.1,

$$(n - k + 1)q^{k-1} = \frac{c_p}{q} \log k + O(1) \geq (c_p + \varepsilon_p) \log k,$$

and the reservoir $M = (1 - o(1))n$ used in Proposition 4.5 still satisfies

$$Mq^{k-1} \geq (c_p + \varepsilon_p/2) \log \log n$$

for all large n . That proposition gives $D_n(p) \geq k - 1$ with high probability.

For the upper neighbor, choose $r_* \in R(\rho)$. Then

$$a_{k+1} := (n - k - 1)q^{k+1} = qc_p \log k + O(1),$$

and $c_p \rho^{r_*} = r_* - 2$. Hence a single layer gives

$$\Phi_{k+1} \geq \binom{k+1}{r_*} F(a_{k+1} \rho^{r_*}) = k^{2+(1-q)(r_*-2)+o(1)} \geq (\log n)^{2+\delta_p}$$

for some $\delta_p > 0$. Also $a_{k+1} \leq (c_p - \varepsilon'_p) \log \log n$, so Proposition 4.2 rules out every shattered $(k+1)$ -set with high probability. Hence $D_n(p) \in \{k-1, k\}$ with high probability, and

$$\mathbb{P}(D_n(p) = k) = \mathbb{P}(Y_k > 0) + o(1) \rightarrow 1 - e^{-e^\tau}.$$

The complementary probability proves the result for $k - 1$. $\square$

6. THE CONTACT SCALE AND THE EXPLICIT CENTER

From this point through Section 10, we return to a sequence $p_n \rightarrow 1/2$ and use the notation $q_n, \rho_n, \theta_n, \alpha_n, a_{n,k}$ introduced in Section 1. The fixed asymmetric parameter from the preceding two sections is no longer in force.

This section develops the deterministic scale $B_k(\theta)$ and proves Theorem 1.5. The estimates are uniform over all sequences $k \rightarrow \infty$ and $\theta = \theta_k \rightarrow 0$.

6.1. Growth and slow variation of the contact scale.

Lemma 6.1. *If $k \rightarrow \infty$ and $\theta = \theta_k \rightarrow 0$, then*

$$(14) \quad \frac{B_k(\theta)}{\log k} \longrightarrow \infty.$$

Moreover,

$$(15) \quad B_k(\theta) \leq B_{k+1}(\theta) \leq B_k(\theta) + \log(k+1),$$

and hence $B_{k+1}(\theta)/B_k(\theta) \rightarrow 1$.

Proof. If $\theta k \leq 2$, take $r = \lfloor k/2 \rfloor$. Then

$$B_k(\theta) \geq e^{-1} \log \binom{k}{\lfloor k/2 \rfloor} = \Theta(k),$$

which implies (14).

Suppose $\theta k > 2$ and take $r = \lfloor 1/\theta \rfloor$. Then $r \leq k/2$, $r \rightarrow \infty$, and $e^{-\theta r} \geq e^{-1}$. The elementary bound $\binom{k}{r} \geq (k/r)^r$ gives

$$B_k(\theta) \geq c\theta^{-1} \log(\theta k)$$

for an absolute $c > 0$. Put $u = \theta^{-1}$. If $u \leq \sqrt{k}$, then

$$\frac{u \log(k/u)}{\log k} \geq \frac{u}{2} \rightarrow \infty.$$

If $u > \sqrt{k}$, then $k/u > 2$ and

$$\frac{u \log(k/u)}{\log k} \geq \frac{u \log 2}{\log k} \rightarrow \infty.$$

This proves (14).

For $0 \leq r \leq k$,

$$\binom{k+1}{r} = \frac{k+1}{k+1-r} \binom{k}{r} \leq (k+1) \binom{k}{r}.$$

Thus

$$e^{-\theta r} \log \binom{k+1}{r} \leq B_k(\theta) + \log(k+1).$$

The additional value $r = k+1$ contributes zero, proving the upper inequality in (15). The lower inequality follows from $\binom{k+1}{r} \geq \binom{k}{r}$. Finally, (14) implies the ratio statement. $\square$

Lemma 6.2. *The moving lattice point in (1) satisfies*

$$K_n = (1 + o(1)) \frac{\log n}{\log 2}.$$

Furthermore,

$$(16) \quad \left(\frac{1}{2} - o(1)\right) B_{K_n+1}(\theta_n) \leq a_{n,K_n+1} < B_{K_n+1}(\theta_n),$$

and

$$(17) \quad B_{K_n}(\theta_n) \leq a_{n,K_n} \leq (2 + o(1)) B_{K_n}(\theta_n).$$

Proof. Let $L = \log n$. Put

$$k_- = \left\lfloor \frac{L - 2 \log L}{\alpha_n} \right\rfloor.$$

Then $a_{n,k_-} = L^{2+o(1)}$, whereas

$$B_{k_-}(\theta_n) \leq \max_r \log \binom{k_-}{r} \leq k_- \log 2 = O(L).$$

Hence $k_- \leq K_n$. On the other hand, for

$$k_+ = \left\lceil \frac{L}{\alpha_n} \right\rceil$$

we have $a_{n,k_+} = O(1)$, while $B_{k_+}(\theta_n) \rightarrow \infty$ by Lemma 6.1. Thus $K_n < k_+$, proving the first assertion.

The sequence $a_{n,k}$ is strictly decreasing in k , while $B_k(\theta_n)$ is nondecreasing. By maximality of K_n ,

$$a_{n,K_n} \geq B_{K_n}(\theta_n), \quad a_{n,K_n+1} < B_{K_n+1}(\theta_n).$$

Moreover,

$$\frac{a_{n,k+1}}{a_{n,k}} = q_n \frac{n-k-1}{n-k} = q_n(1 + o(1))$$

uniformly for $k = O(\log n)$. Combining this relation with Lemma 6.1 gives (16) and (17). $\square$

Lemma 6.3. *Uniformly for integers $k = K_n + O(1)$,*

$$H_{n,k} = \log \binom{n}{k} = (\log 2 + o(1))k^2.$$

In particular, $H_{n,k} = \Theta(k^2)$.

Proof. By Lemma 6.2, $k = (1 + o(1)) \log n / \log 2$ and $k = o(n)$. Stirling's formula, uniformly for such k , gives

$$\log \binom{n}{k} = k \log \frac{n}{k} + k + O(\log k + k^2/n) = k \log n - k \log k + O(k).$$

Since $\log k = o(\log n)$ and $\log n/k \rightarrow \log 2$, division by k^2 proves the assertion. $\square$

6.2. The variational crossover function. Let $\mathcal{C}$ be defined by (4). We collect its elementary analytic properties.

Lemma 6.4. *For every $A \geq 0$, the function $x \mapsto h(x)e^{-Ax}$ has a unique maximizer $x_A \in (0, 1/2]$. For $A > 0$, it is characterized by*

$$\log \frac{1 - x_A}{x_A} = Ah(x_A).$$

Moreover,

$$(18) \quad Ax_A < 1,$$

$$(19) \quad \frac{d}{dA} \log \mathcal{C}(A) = -x_A,$$

$$(20) \quad c \frac{\log(e + A)}{1 + A} \leq \mathcal{C}(A) \leq C \frac{\log(e + A)}{1 + A}$$

for absolute constants $0 < c < C < \infty$. In particular,

$$(21) \quad \mathcal{C}(A) = \frac{\log A}{eA} (1 + o(1)) \quad (A \rightarrow \infty).$$

Finally, there is an absolute $c_0 > 0$ such that

$$(22) \quad x_A \geq \frac{c_0}{1 + A}.$$

Proof. On $(0, 1)$,

$$h'(x) = \log \frac{1 - x}{x}, \quad h''(x) = -\frac{1}{x(1 - x)}.$$

The derivative of $\log h(x) - Ax$ is $h'(x)/h(x) - A$, and

$$\left(\frac{h'}{h} \right)' = \frac{h''h - (h')^2}{h^2} < 0.$$

It decreases from $+\infty$ to $-\infty$, so the maximizer is unique. Since the factor e^{-Ax} favors the left half, $x_A \leq 1/2$, and the critical equation follows.

Using the identity

$$h(x) - xh'(x) = -\log(1 - x) > 0$$

and $A = h'(x_A)/h(x_A)$, we obtain (18). The envelope theorem gives (19).

For the lower bound in (20), first suppose $A \geq 1$ and evaluate at $x = (A + 1)^{-1}$. Since $h(x) \geq x \log(1/x)$ and $Ax \leq 1$,

$$\mathcal{C}(A) \geq e^{-1} \frac{\log(A + 1)}{A + 1}.$$

For $0 \leq A \leq 1$, evaluation at $x = 1/2$ gives a positive absolute lower bound.

For the upper bound it is enough to consider $0 < x \leq 1/2$, where $h(x) \leq x \log(e/x)$. When $A \geq 1$, put $y = Ax$. Then

$$h(x)e^{-Ax} \leq \frac{1}{A} ye^{-y} \log \frac{eA}{y}.$$

The two functions ye^{-y} and $ye^{-y}|\log y|$ are bounded on $(0, \infty)$. Hence the right-hand side is at most $C \log(e + A)/(1 + A)$. The case $A \leq 1$ follows from $h(x) \leq \log 2$. This proves (20).

The maximizer x_A is bounded below by a positive constant when A is bounded. If A is large, then $x_A \leq 1/4$ by the critical equation. For $0 < x \leq 1/4$,

$$h(x) \leq x(\log(1/x) + 1), \quad h'(x) = \log \frac{1-x}{x} \geq \log \frac{1}{2x}.$$

The ratio of the two logarithmic factors is bounded below by a positive absolute constant. Thus

$$A = \frac{h'(x_A)}{h(x_A)} \geq \frac{c}{x_A},$$

which proves (22) after adjusting c to cover bounded A .

For (21), the bounds above imply $x_A \rightarrow 0$. Put $y_A = Ax_A$. Since

$$h(x) = x(\log(1/x) + 1 + o(1)), \quad h'(x) = \log(1/x) + o(1),$$

the critical equation yields

$$y_A = \frac{\log(1/x_A) + o(1)}{\log(1/x_A) + 1 + o(1)} \rightarrow 1.$$

Consequently $\log(1/x_A) = \log A + o(\log A)$ and

$$\mathcal{C}(A) = h(x_A)e^{-Ax_A} = \frac{\log A}{eA}(1 + o(1)).$$

□

Lemma 6.5 (Uniform entropy approximation). *If $k \rightarrow \infty$ and $\theta = \theta_k \rightarrow 0$, then*

$$(23) \quad B_k(\theta) = (1 + o(1))k\mathcal{C}(\theta k).$$

The error is uniform over all such sequences.

Proof. The standard entropy bounds

$$(24) \quad kh(r/k) - \log(k+1) \leq \log \binom{k}{r} \leq kh(r/k)$$

show immediately that

$$B_k(\theta) \leq k\mathcal{C}(\theta k).$$

Let $A = \theta k$ and let x_A maximize $h(x)e^{-Ax}$. Choose an integer r with $|r - kx_A| \leq 1$. By (22),

$$kx_A \geq \frac{c_0 k}{1 + \theta k} \longrightarrow \infty,$$

because $\theta \rightarrow 0$. Also $k(1 - x_A) \geq k/2$. The second derivative of $\log h(x) - Ax$ in a $1/k$ -neighborhood of x_A has magnitude

$$O(x_A^{-2}).$$

Since $kx_A \rightarrow \infty$, we have $1/(k^2 x_A^2) = o(1)$, and Taylor's theorem gives

$$h(r/k)e^{-\theta r} = (1 + o(1))h(x_A)e^{-Ax_A} = (1 + o(1))\mathcal{C}(A).$$

Using the lower bound in (24),

$$B_k(\theta) \geq (1 - o(1))k\mathcal{C}(A) - \log(k+1).$$

By Lemma 6.1, $B_k(\theta) \gg \log k$, so the additive error is negligible relative to $k\mathcal{C}(A)$. This proves (23). $\square$

6.3. Proof of the uniform center theorem. Consider the real equation

$$(25) \quad nq_n^x = x\mathcal{C}(\theta_n x).$$

Lemma 6.6. *Equation (25) has a unique positive solution κ_n . Moreover,*

$$K_n = \kappa_n + O(1).$$

Proof. The left side of (25) is strictly decreasing. For the right side, put $A = \theta_n x$. By (19),

$$\frac{d}{dx} \log(x\mathcal{C}(\theta_n x)) = \frac{1}{x} - \theta_n x_A = \frac{1 - Ax_A}{x} > 0$$

by (18). Thus the right side is strictly increasing, and uniqueness follows. Existence follows from the behavior at zero and infinity.

We first locate the root. Let $L = \log n$ and set

$$x_- = \frac{L - 2 \log L}{\alpha_n}, \quad x_0 = \frac{L}{\alpha_n}.$$

At x_- the left side of (25) is $L^{2+o(1)}$, whereas the right side is at most $x_- \log 2 = O(L)$. At x_0 the left side equals 1, while (20) and $\theta_n = o(1)$ give

$$x_0 \mathcal{C}(\theta_n x_0) \geq c \frac{L \log(e + \theta_n L)}{1 + \theta_n L} \rightarrow \infty.$$

Hence

$$x_- < \kappa_n < x_0, \quad \kappa_n = \Theta(\log n).$$

In particular, Lemma 6.5 is applicable uniformly to integers $k = \kappa_n + O(1)$.

Let

$$R_n(x) = \log n - \alpha_n x - \log x - \log \mathcal{C}(\theta_n x).$$

Then $R_n(\kappa_n) = 0$ and

$$(26) \quad R'_n(x) = -\alpha_n - \frac{1}{x} + \theta_n x \theta_n x < -\alpha_n.$$

By Lemma 6.5, uniformly for integer $k = \kappa_n + O(1)$,

$$\log \frac{a_{n,k}}{B_k(\theta_n)} = R_n(k) + o(1),$$

where replacing $n - k$ by n costs $o(1)$. By (26), $R_n(\kappa_n + c) \leq -c\alpha_n$ and $R_n(\kappa_n - c) \geq c\alpha_n$. Choosing a fixed sufficiently large c determines the sign of the contact inequality and gives $K_n = \kappa_n + O(1)$. $\square$

Proof of Theorem 1.5. Let $L = L_n$, $x_0 = L/\alpha_n$, and $A_n = \theta_n x_0$. Put

$$\delta_n = \log x_0 + \log \mathcal{C}(A_n), \quad m_n = x_0 - \frac{\delta_n}{\alpha_n}.$$

The estimate (23) and Lemma 6.1 imply $x_0 \mathcal{C}(A_n) \rightarrow \infty$, while (20) gives $\delta_n = O(\log L) = o(L)$. Hence

$$\log x_0 - \log m_n = o(1).$$

Also

$$A_n - \theta_n m_n = A_n \frac{\delta_n}{L}.$$

If $A_n \leq 1$, then $|(\log \mathcal{C})'| \leq 1/2$ by (19); if $A_n > 1$, then (18) gives $|(\log \mathcal{C})'(A)| \leq 1/A$. In either case,

$$\log \mathcal{C}(A_n) - \log \mathcal{C}(\theta_n m_n) = o(1).$$

Therefore $R_n(m_n) = o(1)$. By (26),

$$\kappa_n = m_n + o(1).$$

Now apply Lemma 6.6 and then Theorem 1.4. □

Proof of Corollary 1.6. For $A \rightarrow 0$, the envelope identity and the critical equation give

$$x_A = \frac{1}{2} - \frac{\log 2}{4} A + O(A^3)$$

and

$$\log \mathcal{C}(A) = \log \log 2 - \frac{A}{2} + \frac{\log 2}{8} A^2 + O(A^4).$$

Expanding $\alpha_n = \log 2 + \theta_n/2 + O(\theta_n^2)$ in (5) yields parts (i) and (ii).

If $A_n \rightarrow \infty$, use (21) in (5). Since

$$\log A_n = \log \theta_n + \log(L_n/\alpha_n),$$

the terms $\log(L_n/\alpha_n)$ cancel, giving part (iii). □

7. UNIFORM DECORRELATION ACROSS ALL OVERLAPS

We now prove the main probabilistic input. Throughout this section,

$$k = \Theta(\log n), \quad \theta_n \rightarrow 0, \quad B = B_k(\theta_n), \quad a = (n - k)q_n^k.$$

7.1. Contact at the critical level.

Lemma 7.1 (Contact lemma). *Suppose $a < B$ and $\Phi_{n,k} = O(k^2)$. Then*

$$\frac{a}{B} \rightarrow 1.$$

Proof. If not, then along a subsequence $a \leq (1 - \delta)B$ for some fixed $\delta > 0$. Let r_* attain the maximum in the definition of B . Since $F(x) \geq e^{-x}$,

$$\Phi_{n,k} \geq \binom{k}{r_*} e^{-a\rho_n^{r_*}} = \exp\{(B - a)\rho_n^{r_*}\} \geq e^{\delta B}.$$

By Lemma 6.1, $e^{\delta B} \gg k^2$, a contradiction. □

7.2. The pair correction. Let S, T be distinct k -sets, put

$$j = |S \cap T|, \quad d = k - j \in \{1, \dots, k\}.$$

In the Poissonized common outside pool, a compatible pair of cells is specified by the numbers t, b, c of common symbols on $S \cap T$, $S \setminus T$, and $T \setminus S$. Its means are

$$x = a\rho^{t+b}, \quad y = a\rho^{t+c}, \quad z = aq^d \rho^{t+b+c},$$

where, in this subsection, $q = q_n$ and $\rho = \rho_n$. Define

$$\Delta_j = \sum_{t=0}^j \sum_{b,c=0}^d \binom{j}{t} \binom{d}{b} \binom{d}{c} z e^{-x-y+z}.$$

By Lemma 2.4, the logarithmic joint-to-product correction is at most

$$\frac{4}{1 - 2e^{-(1-o(1))a}} \Delta_j = (4 + o(1))\Delta_j.$$

Only the fact that this is $O(\Delta_j)$ will be used.

Lemma 7.2 (Union-exponent gap). *For all admissible t, b, c, d ,*

$$\rho^{t+b} + \rho^{t+c} - q^d \rho^{t+b+c} \geq (2-q)\rho^t.$$

Proof. Assume without loss of generality that $b \geq c$. If $b = c = 0$, the left side equals $\rho^t(2 - q^d) \geq (2 - q)\rho^t$.

Suppose $b \geq 1$. Since $b \leq d$ and $q\rho = 1 - q =: s$,

$$q^d \rho^{b+c} \leq q^b \rho^{b+c} = s^b \rho^c.$$

Therefore

$$\rho^b + \rho^c - q^d \rho^{b+c} \geq \rho^b + \rho^c(1 - s^b) \geq \rho + q.$$

For $q \leq 1/2$,

$$\rho + q - (2 - q) = \frac{(1 - q)(1 - 2q)}{q} \geq 0.$$

Multiplying by ρ^t proves the claim. $\square$

Lemma 7.3 (Uniform pair decorrelation). *Assume*

$$(27) \quad a \geq (1 - o(1))B, \quad a = O(B), \quad \Phi_{n,k} = O(k^2).$$

Then

$$\sup_{0 \leq j \leq k-1} \Delta_j = o(1).$$

The same conclusion holds uniformly when the common-pool intensity is changed by at most $\sqrt{n}k^4$.

Proof. We split according to $d = k - j$. Since $q_n \rightarrow 1/2$, all implicit constants below can be chosen uniformly in n .

High overlap. By Lemma 7.2, summing first over b and c gives

$$\begin{aligned} \Delta_j &\leq aq^d \sum_{t=0}^j \binom{j}{t} \rho^t e^{-(2-q)a\rho^t} \left(\sum_{b=0}^d \binom{d}{b} \rho^b \right)^2 \\ &= aq^{-d} \sum_{t=0}^j \binom{j}{t} \rho^t e^{-(2-q)a\rho^t} \\ (28) \quad &\leq aq^{-d} \sum_{t=0}^k \binom{k}{t} \rho^t e^{-(2-q)a\rho^t}. \end{aligned}$$

By the definition of B ,

$$\log \binom{k}{t} \leq B\rho^t.$$

The first two assumptions in (27) imply that, for some fixed $\delta > 0$,

$$(2 - q)a - B \geq \delta B$$

for all sufficiently large n . Hence each summand in (28) is at most

$$\exp\{\theta t - \delta B e^{\theta t}\}.$$

Since $B \rightarrow \infty$, the function $y \mapsto y - \delta B e^y$ is decreasing on $[0, \infty)$ for all sufficiently large n . Consequently

$$\Delta_j \leq a(k+1)e^{\alpha d - \delta B}, \quad \alpha = \log(1/q).$$

Choose a fixed $c_0 > 0$ so small that $\alpha c_0 < \delta/2$ for all large n . Then

$$(29) \quad \sup_{d \leq c_0 B} \Delta_j = o(1).$$

Smaller overlap. Suppose $d \geq c_0 B$. For compatible cells,

$$\frac{z}{x} \leq s^d, \quad \frac{z}{y} \leq s^d, \quad z \leq \frac{q^d}{a} xy, \quad s = 1 - q.$$

Put $\eta_d = s^d$. Then

$$-x - y + z \leq -(1 - \eta_d/2)(x + y).$$

Extending the compatible-cell sum to all pairs of cells gives

$$(30) \quad \Delta_j \leq \frac{q^d}{a} \left(\sum_{\sigma} x_{\sigma} e^{-(1-\eta_d/2)x_{\sigma}} \right)^2.$$

By [Lemma 6.1](#),

$$\eta_d \leq e^{-cB} = k^{-\omega(1)}.$$

For cells with $x_{\sigma} \leq k^4$,

$$x_{\sigma} e^{-(1-\eta_d/2)x_{\sigma}} \leq 2k^4 e^{-x_{\sigma}}$$

for all sufficiently large n . Their total contribution is $O(k^6)$ because $\sum_{\sigma} e^{-x_{\sigma}} \leq \Phi_{n,k} = O(k^2)$.

For cells with $x_{\sigma} > k^4$,

$$x_{\sigma} e^{-(1-\eta_d/2)x_{\sigma}} \leq e^{-x_{\sigma}/2},$$

and the total is at most $2^k e^{-k^4/2} = o(1)$. Thus the square in (30) is $O(k^{12})$, and

$$\Delta_j \leq \frac{q^d}{a} O(k^{12}) \leq e^{-c'B} k^{12} = o(1).$$

Together with (29), this proves the assertion at the central intensity.

Suppose now that the intensity is changed by at most $\sqrt{n} k^4$. Then a is multiplied by $1 + o(1)$ and remains in (27); in particular the high-overlap gap $(2 - q)a - B \geq \delta B$ persists after decreasing δ . For the smaller-overlap estimate, write $x_{\sigma}(t) = t\pi_{\sigma}$ and split the cells according as $x_{\sigma}(m) \leq 2k^4$ or $x_{\sigma}(m) > 2k^4$, where m is the central intensity. Since $\eta_d \leq e^{-cB}$, we have $\eta_d k^4 = o(1)$. On the first class, uniformly for $|t - m| \leq \sqrt{n} k^4$,

$$x_{\sigma}(t) e^{-(1-\eta_d/2)x_{\sigma}(t)} \leq 3k^4 e^{o(1)} e^{-x_{\sigma}(m)},$$

and the total is $O(k^6)$. On the second class, $x_{\sigma}(t) \geq k^4$ and, for all large n ,

$$x_{\sigma}(t) e^{-(1-\eta_d/2)x_{\sigma}(t)} \leq e^{-x_{\sigma}(t)/2},$$

whose total is $o(1)$. Thus the weighted sum in (30) is uniformly $O(k^6)$ throughout the intensity interval. All preceding estimates are therefore uniform in the stated interval. $\square$

7.3. Fixed-sample joint factorization.

Proposition 7.4 (Joint factorization). *Fix $h \geq 1$. Suppose (27) holds. Then, uniformly over all ordered tuples of distinct k -sets $S_1, \dots, S_h$,*

$$(31) \quad \mathbb{P}(S_1, \dots, S_h \text{ are all externally shattered}) = e^{-h\Phi_{n,k} + o(1)}.$$

The same estimate holds when “externally shattered” is replaced throughout by open-shattered or closed-shattered.

Proof. Let $U = S_1 \cup \dots \cup S_h$ and first use only the $m_0 = n - |U|$ vertices outside U . Poissonize this common pool. When $h = 1$ there is no interpolation term; otherwise apply Lemma 2.4 to the h cylinder partitions. Since $|m_0 - (n - k)| = O_h(k)$, Lemma 2.5 gives a marginal exponent $\Phi_{n,k} + o(1)$ at intensity m_0 . The minimum cell mean is $(1 - o(1))a \rightarrow \infty$. By Lemma 7.3, every pair correction is $o(1)$ uniformly, and there are only $\binom{h}{2}$ pairs. Thus the Poissonized joint probability is

$$Q(m_0) = e^{-h\Phi_{n,k} + o(1)}.$$

The same formula holds uniformly at $m_0 \pm L$, where $L = \sqrt{m_0} k^4$, by the intensity-uniform parts of Lemmas 2.5 and 7.3. Hence $Q(m_0 \pm L) = (1 + o(1))Q(m_0)$.

For the lower-tail condition in Lemma 2.2, a Poisson Chernoff bound gives

$$\mathbb{P}(\text{Poisson}(m_0 - L) > m_0) \leq \exp\{-\Omega(L^2/m_0)\} = e^{-\Omega(k^8)} = o(Q(m_0)),$$

because $\Phi_{n,k} = O(k^2)$. Also $\mathbb{P}(\text{Poisson}(m_0 + L) \geq m_0) = 1 - o(1)$. Therefore Lemma 2.2 yields the same relative estimate with exactly m_0 common outside vertices.

We next restore the vertices in $U \setminus S_i$ as possible witnesses for family i , and, for ordinary shattering, the vertices of S_i itself. Conditional on the graph induced by U , these special vertices supply deterministic trace sets and hence delete at most $C_h k$ cell requirements from each family. Fix any such collection of deleted requirements and let $Q_{\mathcal{I}}(t)$ denote the corresponding Poissonized joint success probability. The interpolation bound in Lemma 2.4 is uniform in the required subcollections. Moreover, uniformly for $|t - m_0| \leq L$, deleting all these cells changes the sum of the marginal logarithms by at most

$$C_h k F((1 - o(1))a) = o(1).$$

Consequently

$$Q_{\mathcal{I}}(m_0 \pm L) = e^{-h\Phi_{n,k} + o(1)} = (1 + o(1))Q_{\mathcal{I}}(m_0)$$

uniformly over all possible deleted collections. Since $Q_{\mathcal{I}}(m_0) \geq Q(m_0) = e^{-O(k^2)}$, the same Poisson-tail estimate is $o(Q_{\mathcal{I}}(m_0))$. The uniform form of Lemma 2.2 therefore applies to every conditional event. Averaging over the graph induced by U proves the external, open, and closed versions of (31). $\square$

8. TWO-POINT CONCENTRATION AND THE CRITICAL LAW

We now prove Theorems 1.4 and 8.3. Recall that K_n is the largest integer for which $a_{n,k} \geq B_k(\theta_n)$.

8.1. The lower contact level.

Proposition 8.1. *With high probability, there is an externally shattered K_n -set. Consequently,*

$$D_n^{\circ}, D_n^{\text{cl}} \geq K_n \quad \text{with high probability.}$$

Proof. Put $k = K_n$, $a = a_{n,k}$ and $B = B_k(\theta_n)$. Since $a \geq B$, for every r ,

$$\binom{k}{r} e^{-a\rho_n^r} \leq 1.$$

Also $a \geq B \gg \log k$. Therefore

$$\Phi_{n,k} \leq \frac{1}{1 - e^{-a}} \sum_{r=0}^k \binom{k}{r} e^{-a\rho_n^r} \leq (1 + o(1))(k + 1) = O(k).$$

By (17), $a = O(B)$, so Proposition 7.4 applies.

Let Y count externally shattered k -sets. Then

$$\mathbb{E}Y = \binom{n}{k} e^{-\Phi_{n,k} + o(1)} = \exp\{H_{n,k} - O(k)\} \rightarrow \infty,$$

by Lemma 6.3. Moreover,

$$\mathbb{E}[Y(Y-1)] = (1 + o(1)) \binom{n}{k}^2 e^{-2\Phi_{n,k}} = (1 + o(1))(\mathbb{E}Y)^2.$$

Thus $\text{Var } Y = o((\mathbb{E}Y)^2)$, and the second-moment method gives $Y > 0$ with high probability. $\square$

8.2. The level two steps above contact.

Proposition 8.2. *With high probability, no $(K_n + 2)$ -set is open- or closed-shattered. Consequently,*

$$D_n^{\circ}, D_n^{\text{cl}} \leq K_n + 1 \quad \text{with high probability.}$$

Proof. Put $k = K_n + 2$, $a = a_{n,k}$ and $B = B_k(\theta_n)$. Since $K_n + 1$ fails the contact inequality,

$$a_{n,K_n+1} < B_{K_n+1}(\theta_n).$$

Using

$$\frac{a_{n,k}}{a_{n,k-1}} = q_n(1 + o(1))$$

and the slow variation in (15), we obtain

$$(32) \quad a \leq \left(\frac{1}{2} + o(1)\right) B.$$

Choose r_* attaining B . In the Poissonized outside exposure, the number Z_* of empty cells in this Hamming layer is binomial with mean

$$\mu_* = \binom{k}{r_*} e^{-a\rho_n^{r_*}} = \exp\{(B - a)\rho_n^{r_*}\}.$$

By (32),

$$\log \mu_* \geq \left(\frac{1}{2} - o(1)\right) B \rho_n^{r_*} \geq \left(\frac{1}{2} - o(1)\right) B.$$

Since $B/\log k \rightarrow \infty$, we have $\mu_* = k^{\omega(1)}$, in particular $\mu_* \gg k^3$ and $\mu_* \gg H_{n,k}$.

If a fixed k -set is open- or closed-shattered, its k internal vertices can supply at most k externally missing patterns. Thus shattering implies $Z_* \leq k$. A Chernoff lower-tail bound in the Poissonized model gives

$$\mathbb{P}(Z_* \leq k) \leq e^{-c\mu_*}$$

for an absolute $c > 0$. The event $\{Z_* \leq k\}$ is increasing in the number of outside traces, so its probability with exactly $n - k$ traces is at most $2 + o(1)$ times the Poissonized probability. A union bound yields

$$\mathbb{P}(\exists \text{ shattered } k\text{-set}) \leq \exp\{H_{n,k} - c'\mu_*\} = o(1).$$

$\square$

8.3. Critical counts. Let $Y_{n,k}^{\text{ext}}, Y_{n,k}^{\circ}, Y_{n,k}^{\text{cl}}$ be the numbers of externally, open-, and closed-shattered k -sets, respectively.

Theorem 8.3. Assume $k = K_n + 1$ and $\Phi_{n,k} = O(k^2)$. For every fixed $h \geq 1$ and $\star \in \{\text{ext}, \circ, \text{cl}\}$,

$$\mathbb{E}[(Y_{n,k}^{\star})_h] = (1 + o(1))\lambda_n^h.$$

If $\sup_n \lambda_n < \infty$, then

$$d_{\text{TV}}(\mathcal{L}(Y_{n,k}^{\star}), \text{Poisson}(\lambda_n)) \rightarrow 0.$$

If $\lambda_n \rightarrow \infty$, then $Y_{n,k}^{\star}/\lambda_n \rightarrow 1$ in probability. In particular, $H_{n,k} - \Phi_{n,k} \rightarrow \tau$ implies convergence to a Poisson random variable of mean e^{τ} and the lattice probability $1 - e^{-e^{\tau}}$.

Fix from now on

$$k = K_n + 1, \quad H = H_{n,k}, \quad \Phi = \Phi_{n,k}, \quad \lambda = e^{H-\Phi}.$$

Proposition 8.4. If $\Phi = O(k^2)$, then for every fixed $h \geq 1$ and every $\star \in \{\text{ext}, \circ, \text{cl}\}$,

$$\mathbb{E}[(Y_{n,k}^{\star})_h] = (1 + o(1))\lambda^h.$$

Proof. By (16), $a \geq (1/2 - o(1))B$ and $a < B$. Since $\Phi = O(k^2)$, Lemma 7.1 improves this to $a = (1 - o(1))B$. Thus Proposition 7.4 applies.

The number of ordered h -tuples of distinct k -sets is

$$(1 + o(1)) \binom{n}{k}^h,$$

because h is fixed and $\binom{n}{k} \rightarrow \infty$. Hence

$$\mathbb{E}[(Y_{n,k}^{\star})_h] = (1 + o(1)) \binom{n}{k}^h e^{-h\Phi} = (1 + o(1))e^{h(H-\Phi)}.$$

□

Proof of Theorem 8.3. The factorial-moment assertion is Proposition 8.4.

Suppose first that $\sup_n \lambda_n < \infty$. If the claimed total-variation convergence failed, there would be a subsequence on which the distance is bounded below. Passing to a further subsequence, assume $\lambda_n \rightarrow \lambda \in [0, \infty)$. If $\lambda > 0$, the factorial moments converge to those of $\text{Poisson}(\lambda)$, so the standard factorial-moment criterion [2] gives weak convergence to $\text{Poisson}(\lambda)$. If $\lambda = 0$, the first moment tends to zero and hence $Y_{n,k}^{\star} \rightarrow 0$ in probability. In either case, on the countable state space $\mathbb{N} \cup \{0\}$, pointwise convergence of the probability masses and conservation of total mass imply convergence in total variation to $\text{Poisson}(\lambda)$. Finally, using the standard coupling bound

$$d_{\text{TV}}(\text{Poisson}(u), \text{Poisson}(v)) \leq 1 - e^{-|u-v|} \leq |u-v|,$$

we have $d_{\text{TV}}(\text{Poisson}(\lambda_n), \text{Poisson}(\lambda)) \rightarrow 0$. The triangle inequality contradicts the choice of the subsequence and proves part (i).

If $\lambda_n \rightarrow \infty$, then

$$\mathbb{E}Y_{n,k}^{\star} = (1 + o(1))\lambda_n$$

and

$$\mathbb{E}[(Y_{n,k}^{\star})_2] = (1 + o(1))\lambda_n^2.$$

Thus

$$\text{Var } Y_{n,k}^{\star} = \lambda_n + o(\lambda_n^2),$$

and Chebyshev's inequality proves part (ii). The final assertions follow by taking $\lambda_n \rightarrow e^{\tau}$. □

8.4. Proof of the complete law.

Proof of Theorem 1.4. By Propositions 8.1 and 8.2, both dimensions lie in $\{K_n, K_n + 1\}$ with high probability.

We argue by the subsequence principle. Along any subsequence, pass to a further subsequence on which one of the following alternatives holds.

First suppose $\lambda_n \rightarrow \lambda \in (0, \infty)$. Then $\Phi = H + O(1) = O(k^2)$, and Theorem 8.3 gives

$$\mathbb{P}(Y_{n,k}^{\text{ext}} = 0) = e^{-\lambda_n} + o(1).$$

By Lemma 2.6, the expected number of open- or closed-shattered but not externally shattered k -sets is

$$\binom{n}{k} o(e^{-\Phi}) = o(\lambda_n) = o(1).$$

Thus all three notions agree at the critical level with probability $1 - o(1)$, proving (2)–(3) in this case.

If $\lambda_n \rightarrow \infty$, then $H - \Phi \rightarrow \infty$, so $\Phi = O(k^2)$ and Theorem 8.3 gives $Y_{n,k}^{\text{ext}} > 0$ with high probability. Hence both dimensions equal $K_n + 1$ with high probability, while $1 - e^{-\lambda_n} \rightarrow 1$.

Finally suppose $\lambda_n \rightarrow 0$. If $\Phi = O(k^2)$, then Lemma 7.1 and Eq. (16) give $a = (1 - o(1))B$. Thus Proposition 7.4 with $h = 1$ and a union bound give

$$\mathbb{P}(\exists \text{ ordinary shattered } k\text{-set}) \leq e^{H - \Phi + o(1)} = o(1).$$

If $\Phi/k^2 \rightarrow \infty$ along a further subsequence, then $a \geq (1/2 - o(1))B \gg \log k$ and Lemma 2.7 applies: indeed $H = \Theta(k^2)$, while

$$k \log \left(e + \frac{\Phi}{k} \right) = O(k^2) = o(\Phi),$$

because $a \rightarrow \infty$ implies $\Phi \leq 2^k F(a) \leq 2^k$ for all sufficiently large n , while $\Phi/k^2 \rightarrow \infty$. Thus no ordinary shattered critical set exists with high probability. Since $1 - e^{-\lambda_n} \rightarrow 0$, the desired formula follows.

Every subsequence has a further subsequence covered by one of these three alternatives. The subsequence principle therefore proves the theorem without any convergence assumption on λ_n . The equality $D_n^\circ = D_n^{\text{cl}}$ follows in each case. $\square$

9. THE NEAR-SYMMETRIC REGIME

The global crossover from the symmetric to the asymmetric mechanism is encoded by the saddle-point behavior of $\Phi_k(p_n)$. As an immediate consequence of Theorem 1.5, if $p_n = 1/2 + \varepsilon_n$ and $|\varepsilon_n| \log n = O(1)$, then

$$D_n^\circ(p_n), D_n^{\text{cl}}(p_n) = \log_2 n - \log_2 \log n + O_{\mathbb{P}}(1).$$

The sharper question is when the two symmetric lattice points themselves persist; this is governed by the exact equation $\Phi_{n,k} \approx H_{n,k}$.

The Hamming-layer average gives a substantially wider range in which the two symmetric lattice points themselves persist. Put

$$L = \log n, \quad \eta_n = |\varepsilon_n|, \quad \ell_n = \log_2(n/L), \quad J_n = \lfloor \ell_n \rfloor.$$

For a word $\sigma \in \{0, 1\}^k$, write

$$\pi_n(\sigma) = p_n^{|\sigma|} (1 - p_n)^{k - |\sigma|}, \quad \Phi_{n,k} = \sum_{\sigma} F((n - k)\pi_n(\sigma)), \quad H_{n,k} = \log \binom{n}{k}.$$

Lemma 9.1. *Let $k = J_n + 1$, assume $\eta_n k = o(1)$, and suppose $\Phi_{n,k} = O(L^2)$. For every fixed h and every ordered tuple of distinct k -sets $S_1, \dots, S_h$,*

$$(33) \quad \mathbb{P}(S_1, \dots, S_h \text{ are all externally shattered}) = \exp\{-h\Phi_{n,k} + o(1)\},$$

uniformly over the tuple. The same relative estimate holds for open or closed shattering. In particular, uniformly over k -sets S and for either ordinary convention,

$$\mathbb{P}(S \text{ is ordinarily but not externally shattered}) = o(e^{-\Phi_{n,k}}).$$

Proof. Put $B = 2^k$, $m = n - k$, $u = m/B$, and

$$r_- = (1 - 2\eta_n)^k, \quad r_+ = (1 + 2\eta_n)^k.$$

Every cell mean lies in $[ur_-, ur_+]$, and $r_{\pm} = 1 + o(1)$. Since $F(x) \geq e^{-x}$ and $B \in (n/L, 2n/L]$, the assumption $\Phi_{n,k} = O(L^2)$ implies

$$ur_+ \geq L - O(\log L), \quad ur_- = (1 - o(1))L.$$

Let $U = S_1 \cup \dots \cup S_h$, $m_0 = n - |U|$, and $u_0 = m_0/B$. Then $m_0 = m - O_h(k)$ and $u_0 = u + o(1)$. For two compatible cylinders belonging to distinct k -sets, conditioning on either cylinder leaves at least one new Bernoulli coordinate. With $q_n = \min(p_n, 1 - p_n)$, their union therefore has measure at least $(1 + q_n)$ times the minimum k -cell probability. Hence each pair correction in Lemma 2.4 is at most

$$m_0 e^{-(1+q_n)u_0 r_-} = n^{-1/2+o(1)} = o(1),$$

because the compatible intersections partition the atom space and their means sum to m_0 . Using first only the vertices outside U changes every marginal exponent by $o(1)$. We give the uniform derivative bound that also supports the de-Poissonization below. For every fixed $A > 0$, uniformly when

$$|t - m| \leq A\sqrt{n}L + O_h(k),$$

put $x_{\sigma}(t) = t\pi_n(\sigma)$. Since every cell mean at $t = m$ is $O(L)$,

$$\max_{\sigma} x_{\sigma}(t) = O(L), \quad \max_{\sigma} |x_{\sigma}(t) - x_{\sigma}(m)| = O(L^2/\sqrt{n}) + o(1) = o(1).$$

The hypothesis $\Phi_{n,k} = O(L^2)$ and $F(x) \geq e^{-x}$ give

$$\sum_{\sigma} e^{-x_{\sigma}(t)} \leq e^{o(1)} \sum_{\sigma} e^{-x_{\sigma}(m)} = O(L^2).$$

Moreover, $\min_{\sigma} x_{\sigma}(t) \rightarrow \infty$, so $1 - e^{-x_{\sigma}(t)} \geq 1/2$ for all large n . Consequently

$$\left| \frac{d}{dt} \sum_{\sigma} F(t\pi_n(\sigma)) \right| \leq \frac{2}{t} \sum_{\sigma} x_{\sigma}(t) e^{-x_{\sigma}(t)} = O(L^3/n).$$

The multi-partition lemma now gives the Poissonized version of (33).

Choose $T = A\sqrt{n}L$ with A sufficiently large. The last derivative bound makes the exponent stable by $o(1)$ for intensities differing by at most T , while the Poisson tail is $e^{-\Omega(A^2 L^2)} = o(e^{-h\Phi_{n,k}})$. Lemma 2.2 gives relative de-Poissonization.

Finally condition on the graph induced by the union of the candidate sets. The vertices in the union, including internal vertices for ordinary open or closed shattering, delete only $O_h(k)$ requirements. Uniformly over those requirements the logarithmic change is at most

$$O_h(k)F(ur_-) = o(1),$$

and Lemma 2.4 is uniform in deleted cells. This proves the stated external and ordinary versions. Since external shattering implies ordinary shattering, subtracting the corresponding one-set probabilities gives the final estimate. $\square$

Theorem 9.2. *Suppose*

$$(34) \quad \varepsilon_n^2(\log n)^3 = o(\log \log n).$$

Then both neighborhood dimensions lie in $\{J_n, J_n + 1\}$ with high probability. At $k = J_n + 1$, the exact exponent gives the following three alternatives:

- (i) *if $\Phi_{n,k} - H_{n,k} \rightarrow +\infty$, then $D_n^\circ = D_n^{\text{cl}} = J_n$ with high probability;*
- (ii) *if $H_{n,k} - \Phi_{n,k} \rightarrow +\infty$, then $D_n^\circ = D_n^{\text{cl}} = J_n + 1$ with high probability;*
- (iii) *if $H_{n,k} - \Phi_{n,k} \rightarrow \tau \in \mathbb{R}$, then*

$$\mathbb{P}(D_n^\circ = D_n^{\text{cl}} = J_n + 1) \rightarrow 1 - e^{-e^\tau}.$$

Proof of Theorem 9.2. First take $k = J_n$. Write $u = (n - k)/2^k \in [L - o(1), 2L + o(L)]$ and $R_\sigma = 2^k \pi_n(\sigma)$. Under the uniform distribution on words, the multiset of the R_σ is the law of

$$R = \prod_{i=1}^k (1 + 2\eta_n \xi_i), \quad \mathbb{P}(\xi_i = 1) = \mathbb{P}(\xi_i = -1) = 1/2.$$

Writing $Y = \log R$ and using $e^Y - 1 \geq Y$ gives

$$(35) \quad \sum_{\sigma} e^{-u R_\sigma} \leq 2^k e^{-u} \mathbb{E} e^{-u Y} = 2^k e^{-u} \left(\frac{(1 + 2\eta_n)^{-u} + (1 - 2\eta_n)^{-u}}{2} \right)^k.$$

Condition (34) implies $\eta_n L = o(1)$, and Taylor expansion shows that the logarithm of the last factor is $O(\eta_n^2 u^2 + \eta_n^2 u)$. After multiplication by $k = O(L)$ this is $o(\log L)$. Since $2^k e^{-u} \leq (1 + o(1))L^{-1}$, the right side of (35) is $L^{-1+o(1)} = o(1)$. A union bound over the missing traces shows that this fixed J_n -set is externally shattered with high probability.

For $k = J_n + 2$, one has $2^k > 2n/L$ and $(1 + 2\eta_n)^k = 1 + o(1)$. Thus every Poissonized cell mean is at most $(1/2 + o(1))L$, and the expected number of empty cells is at least $n^{1/2-o(1)}$. Chernoff's inequality, monotonicity back to the fixed sample, and a union bound over $e^{O(L^2)}$ candidate sets show that no such set is open- or closed-shattered with high probability. This proves the two-point assertion.

Return to $k = J_n + 1$, and put $H = H_{n,k}$ and $\Phi = \Phi_{n,k}$. Let Y_k count externally shattered $(J_n + 1)$ -sets. If $H - \Phi$ is bounded below, Lemma 9.1 gives, for every fixed h ,

$$\mathbb{E}[(Y_k)_h] = \left(\binom{n}{k} \right)_h e^{-h\Phi+o(1)}.$$

This proves the Poisson limit when $H - \Phi \rightarrow \tau$ and the second-moment lower bound when $H - \Phi \rightarrow +\infty$. In the critical case, the ordinary open and closed counts agree with the external count with probability $1 - o(1)$: for either convention, the expected number of discrepancies is

$$\binom{n}{k} o(e^{-\Phi}) = o(e^{H-\Phi}) = o(1)$$

by Lemma 9.1. In the divergent lower regime, the existence of an externally shattered set already proves the required conclusion for both conventions.

If $\Phi - H \rightarrow +\infty$, condition on the internal graph of a fixed candidate. Its internal vertices delete at most k requirements. Here the minimum outside mean is at least $(1/2 - o(1))L$, so the Poissonized shattering probability is at most $e^{-\Phi+o(1)}$. Monotonicity costs only a factor $2 + o(1)$ for the fixed outside sample, and a union bound gives $e^{H-\Phi+o(1)} = o(1)$. The three alternatives follow. $\square$

The exact exponent can be expanded at the first scale at which the symmetric limiting law changes.

Theorem 9.3. *At $k = J_n + 1$, put $M = 2^k$, $u = (n - k)/M$, $\lambda = Me^{-u}$, and $H = \log \binom{n}{k}$. If, along a subsequence,*

$$H - \lambda \rightarrow \tau, \quad \varepsilon_n (\log n)^{5/2} \rightarrow c,$$

then

$$\mathbb{P}(D_n^\circ = D_n^{\text{cl}} = J_n + 1) \longrightarrow 1 - \exp \left\{ -\exp \left(\tau - \frac{2c^2}{(\log 2)^2} \right) \right\}.$$

Proof of Theorem 9.3. Write $\Phi = \Phi_{n,k}$. Let $R_\sigma = 2^k \pi_n(\sigma) = 1 + \delta_\sigma$. At a symmetric critical subsequence,

$$\lambda \sim H \sim L^2 / \log 2, \quad u \sim L, \quad k \sim L / \log 2.$$

The scale assumption gives

$$\max_\sigma |\delta_\sigma| = O(|\varepsilon_n|k) = O(L^{-3/2}), \quad u \max_\sigma |\delta_\sigma| = o(1).$$

Furthermore

$$\frac{1}{2^k} \sum_\sigma \delta_\sigma = 0, \quad \frac{1}{2^k} \sum_\sigma \delta_\sigma^2 = (1 + 4\varepsilon_n^2)^k - 1 = 4k\varepsilon_n^2(1 + o(1)).$$

For $x > 0$,

$$0 \leq F(x) - e^{-x} \leq \frac{e^{-2x}}{1 - e^{-x}}.$$

Since $uR_\sigma = u + o(1)$ uniformly,

$$\sum_\sigma \{F(uR_\sigma) - e^{-uR_\sigma}\} \leq (1 + o(1))2^k e^{-2u} = O(\lambda^2/2^k) = o(1).$$

Moreover,

$$\frac{1}{2^k} \sum_\sigma |\delta_\sigma|^3 \leq \max_\sigma |\delta_\sigma| \frac{1}{2^k} \sum_\sigma \delta_\sigma^2 = O(|\varepsilon_n|^3 k^2).$$

Taylor expansion and the linear cancellation therefore give

$$\Phi_{n,k} = \lambda + 2\lambda u^2 k \varepsilon_n^2 + o(1).$$

Here the cubic error is $O(\lambda u^3 |\varepsilon_n|^3 k^2) = o(1)$. Finally,

$$2\lambda u^2 k \varepsilon_n^2 \longrightarrow \frac{2c^2}{(\log 2)^2}.$$

Condition (34) holds, so apply Theorem 9.2 with $H - \Phi \rightarrow \tau - 2c^2/(\log 2)^2$. $\square$

In particular, if $|\varepsilon_n|(\log n)^{5/2} = o(1)$, the full symmetric two-point law is stable. Indeed, when $\lambda \leq 2H$ the preceding expansion gives $\Phi_{n,k} = \lambda + o(1)$. In the remaining upper regime, convexity of F and $2^{-k} \sum_\sigma R_\sigma = 1$ give

$$\Phi_{n,k} = 2^k \frac{1}{2^k} \sum_\sigma F(uR_\sigma) \geq 2^k F(u) \geq 2^k e^{-u} = \lambda.$$

Thus the three alternatives of Theorem 1.1 continue to hold, jointly for the open and closed conventions.

10. THE MICROSCOPIC CONTACT WINDOW

The exact exponent $\Phi_{n,k}$ is the natural formulation of the critical law. This section gives a deterministic calibration of the very thin parameter interval in which $H_{n,k} - \Phi_{n,k} = O(1)$.

Fix $k = K_n + 1$, and abbreviate

$$q = q_n, \quad \rho = \rho_n = e^\theta, \quad B = B_k(\theta).$$

Choose

$$r_* \in \arg \max_{0 \leq r \leq k} \rho^{-r} \log \binom{k}{r}$$

and put

$$L_* = \log \binom{k}{r_*} = B\rho^{r_*}.$$

The endpoints $r = 0, k$ contribute zero, so $1 \leq r_* \leq k - 1$ for all large k .

10.1. Contact weights and near-contact rigidity. Define

$$u_r = \binom{k}{r} e^{-B\rho^r}, \quad \Xi_{k,\theta} = \sum_{r=0}^k u_r.$$

Then $0 < u_r \leq 1$ and $u_{r_*} = 1$.

Lemma 10.1 (Contact partition bound). *There is an absolute constant C such that*

$$1 \leq \Xi_{k,\theta} \leq C\sqrt{k}.$$

Proof. Let

$$f_r = \log \binom{k}{r} - B\rho^r.$$

Then $f_r \leq 0$ and $f_{r_*} = 0$. For $1 \leq r \leq k - 1$,

$$\begin{aligned} \Delta^2 f_r &= \log \frac{r(k-r)}{(r+1)(k-r+1)} - B\rho^{r-1}(\rho-1)^2 \\ &\leq -\frac{1}{r+1} - \frac{1}{k-r+1} \leq -\frac{4}{k+2}. \end{aligned}$$

Since r_* is a discrete maximum, summing the first-difference inequalities on either side gives

$$f_{r_* \pm m} \leq -\frac{cm(m-1)}{k}$$

for an absolute $c > 0$. Therefore

$$\Xi_{k,\theta} \leq 1 + 2 \sum_{m \geq 1} e^{-cm(m-1)/k} \leq C\sqrt{k}.$$

□

For $0 \leq r \leq k$, set

$$b_r = \rho^{-r} \log \binom{k}{r}, \quad B = \max_r b_r.$$

Lemma 10.2 (Near-contact rigidity). *Let $k \rightarrow \infty$ and $\theta = \theta_k \rightarrow 0$. If $r = r_k$ satisfies*

$$b_r \geq (1 - o(1))B,$$

then, for any maximizing layer r_ ,*

$$\theta|r - r_*| \rightarrow 0.$$

The conclusion is uniform: for every $\eta_k \rightarrow 0$,

$$\sup_{\{r: b_r \geq (1 - \eta_k)B\}} \theta|r - r_*| \rightarrow 0.$$

Proof. Put $A_k = \theta k$. We prove the sequential assertion; the uniform form then follows by contradiction and passage to a subsequence.

Suppose first that (A_k) is bounded. Passing to a subsequence, let $A_k \rightarrow A \in [0, \infty)$. Write $x = r/k$. The upper entropy bound and Lemma 6.5 give

$$\frac{b_r}{B} \leq \frac{h(x)e^{-A_k x}}{\mathcal{C}(A_k)}(1 + o(1)).$$

Thus near-maximality of b_r forces $h(x)e^{-A_k x}/\mathcal{C}(A_k) \rightarrow 1$. By compactness of $[0, 1]$ and uniqueness of the maximizer of $h(x)e^{-Ax}$, we obtain $r/k \rightarrow x_A$. Applying the same argument to a maximizing layer gives $r_*/k \rightarrow x_A$, and hence

$$\theta(r - r_*) = A_k \left(\frac{r}{k} - \frac{r_*}{k} \right) \rightarrow 0.$$

Now suppose $A_k \rightarrow \infty$. Since

$$B = (1 + o(1)) \frac{\log A_k}{e\theta}$$

by Lemmas 6.4 and 6.5, a near-maximizing layer satisfies $r = o(k)$: if $r/k \geq \varepsilon > 0$, then $b_r \leq k \log 2 e^{-\varepsilon A_k} = o(B)$. It also satisfies $r \rightarrow \infty$, because bounded r gives $b_r = O(\log k) = o(B)$. Set $y = \theta r \in (0, A_k]$. The global binomial estimate $\binom{k}{r} \leq (ek/r)^r$ yields

$$(36) \quad \frac{b_r}{B} \leq (1 + o(1)) e y e^{-y} \frac{\log(eA_k/y)}{\log A_k}.$$

We first exclude the two ends of the y -range using only this upper bound. If $y \rightarrow \infty$, then $\log(eA_k/y) \leq \log(eA_k)$ and the right side of (36) tends to zero. If $y \rightarrow 0$ and $y \geq A_k^{-1/2}$, the logarithmic ratio is at most $3/2 + o(1)$, so the right side again tends to zero. If $0 < y < A_k^{-1/2}$, the function $y \mapsto y \log(eA_k/y)$ is increasing on this interval, and therefore

$$y \frac{\log(eA_k/y)}{\log A_k} = O(A_k^{-1/2}) = o(1).$$

Hence near-maximality forces y to remain in a fixed compact subset of $(0, \infty)$. On every such compact subset the logarithmic ratio in (36) converges uniformly to one. Since $e y e^{-y} \leq 1$, with equality only at $y = 1$, near-maximality implies $\theta r = y \rightarrow 1$. The same argument applied to r_* gives $\theta r_* \rightarrow 1$, and the conclusion follows.

If the uniform assertion failed for some $\eta_k \rightarrow 0$, a subsequence of layers with $b_r \geq (1 - \eta_k)B$ and $\theta|r - r_*|$ bounded away from zero would contradict one of the two arguments above. $\square$

10.2. The tilted contact partition. For $T \geq 0$, define

$$Z_{k,\theta}(T) = \sum_{r=0}^k u_r e^{T\rho^{r-r_*}}.$$

If

$$a(T) = B - \rho^{-r_*}T,$$

then the first-order missing-pattern sum has the exact representation

$$\Lambda_k(a(T), \theta) := \sum_{r=0}^k \binom{k}{r} e^{-a(T)\rho^r} = Z_{k,\theta}(T).$$

Lemma 10.3 (Tilted moment rigidity). *Uniformly for $0 \leq T \leq 3 \log k$ and every fixed integer $j \geq 0$,*

$$(37) \quad \sum_{r=0}^k \rho^{j(r-r_*)} u_r e^{T\rho^{r-r_*}} = (1 + o(1)) Z_{k,\theta}(T).$$

Moreover, uniformly in this range,

$$(38) \quad \Phi_{k,\theta}(T) := \sum_{r=0}^k \binom{k}{r} F(a(T)\rho^r) = Z_{k,\theta}(T) + o(1).$$

Proof. Set

$$\eta_k = \sqrt{\frac{\log k}{B}}, \quad \mathcal{N} = \{r : b_r \geq (1 - \eta_k)B\}.$$

Then $\eta_k \rightarrow 0$ and $\eta_k B = \sqrt{B \log k} \gg \log k$. Write $\delta = T/L_*$. Since $T \leq 3 \log k$ and $L_* \geq B$, we have $\delta = o(\eta_k)$ uniformly. For $r \notin \mathcal{N}$,

$$\log(u_r e^{T\rho^{r-r_*}}) = (b_r - (1 - \delta)B)\rho^r \leq -\frac{1}{2}\eta_k B \rho^r.$$

After multiplication by $\rho^{j(r-r_*)}$, the logarithm of the r th summand is at most

$$-\frac{1}{2}\eta_k B e^{\theta r} + j\theta r - j\theta r_*.$$

The function $y \mapsto -\eta_k B e^y/2 + jy$ is decreasing on $[0, \infty)$ for all large k . Hence every such summand is at most $e^{-\eta_k B/2}$, and therefore

$$(39) \quad \sum_{r \notin \mathcal{N}} \rho^{j(r-r_*)} u_r e^{T\rho^{r-r_*}} \leq (k+1)e^{-\eta_k B/2} = o(1)$$

uniformly in T .

On $\mathcal{N}$, [Lemma 10.2](#) gives $\rho^{j(r-r_*)} = 1 + o(1)$ uniformly. Let $W_{\mathcal{N}}(T) = \sum_{r \in \mathcal{N}} u_r e^{T\rho^{r-r_*}}$. The term $r = r_*$ shows $W_{\mathcal{N}}(T) \geq e^T \geq 1$. Thus (39) gives

$$Z(T) = W_{\mathcal{N}}(T) + o(1)$$

and

$$\sum_r \rho^{j(r-r_*)} u_r e^{T\rho^{r-r_*}} = (1 + o(1))W_{\mathcal{N}}(T) + o(1) = (1 + o(1))Z(T),$$

proving (37).

The same rigidity gives $\rho^{r-r_*} = 1 + o(1)$ uniformly on $\mathcal{N}$. Since $u_r \leq 1$ and $T \leq 3 \log k$,

$$Z(T) \leq k^{4+o(1)}.$$

Moreover, $a(T) \geq B - 3 \log k \sim B \gg \log k$. Using

$$0 \leq F(x) - e^{-x} \leq \frac{e^{-2x}}{1 - e^{-x}},$$

we obtain uniformly

$$0 \leq \Phi_{k,\theta}(T) - Z_{k,\theta}(T) \leq \frac{e^{-a(T)}}{1 - e^{-a(T)}} Z_{k,\theta}(T) = o(1),$$

which proves (38). $\square$

10.3. The exact microscopic center. Let $H = H_{n,k}$. By Lemma 10.1,

$$Z(0) = \Xi_{k,\theta} = O(\sqrt{k}) = o(H),$$

whereas the r_* term gives

$$Z(3 \log k) \geq k^3 > H$$

for all sufficiently large n , by Lemma 6.3. Thus there is a unique

$$\hat{T} \in (0, 3 \log k) \quad \text{such that} \quad Z_{k,\theta}(\hat{T}) = H.$$

Lemma 10.4. *Uniformly for $0 \leq T \leq 3 \log k$,*

$$(40) \quad \Phi'_{k,\theta}(T) = (1 + o(1)) Z_{k,\theta}(T),$$

$$(41) \quad \Phi''_{k,\theta}(T) = (1 + o(1)) Z_{k,\theta}(T).$$

If T_c is the unique solution of $\Phi_{k,\theta}(T_c) = H$, then

$$(42) \quad T_c = \hat{T} + o(H^{-1}).$$

Proof. Since $a'(T) = -\rho^{-r_*}$,

$$\Phi'(T) = \sum_{r=0}^k \binom{k}{r} \frac{\rho^{r-r_*}}{e^{a(T)\rho^r} - 1}.$$

Uniformly in T , $a(T) \rightarrow \infty$ and $(e^x - 1)^{-1} = e^{-x}(1 + O(e^{-a(T)}))$. Thus (40) follows from Lemma 10.3 with $j = 1$. Differentiating again gives

$$\Phi''(T) = \sum_{r=0}^k \binom{k}{r} \rho^{2(r-r_*)} \frac{e^{a(T)\rho^r}}{(e^{a(T)\rho^r} - 1)^2},$$

and the estimate $e^x/(e^x - 1)^2 = e^{-x}(1 + O(e^{-a(T)}))$, together with the case $j = 2$, proves (41). The same $j = 1$ estimate gives

$$(43) \quad \frac{Z'(T)}{Z(T)} = 1 + o(1)$$

uniformly on $[0, 3 \log k]$.

By (38), $\Phi(0) = O(\sqrt{k}) = o(H)$, whereas $\Phi(3 \log k) \geq Z(3 \log k) \geq k^3 > H$. Hence the strictly increasing function Φ has a unique solution $T_c \in (0, 3 \log k)$ of $\Phi(T_c) = H$. We next localize it. The identity $Z(\hat{T}) = H$ and the r_* term imply $\hat{T} \leq \log H = 2 \log k + O(1)$; furthermore (43) on every fixed interval starting at zero, together with $Z(0) = O(\sqrt{k}) = o(H)$, shows $\hat{T} \rightarrow \infty$. Thus, for every fixed $c > 0$, both $\hat{T} - c$ and $\hat{T} + c$ lie in $(0, 3 \log k)$ for all large k . Integrating (43) gives

$$Z(\hat{T} - c) \leq e^{-c/2} H, \quad Z(\hat{T} + c) \geq e^{c/2} H$$

for all sufficiently large k . Since $\Phi(T) = Z(T) + o(1)$ uniformly, these inequalities imply $T_c \in (\hat{T} - c, \hat{T} + c)$. On this fixed neighborhood, $Z(T) = \Theta(H)$ and hence (40) gives $\Phi'(T) = \Theta(H)$. Finally, $\Phi(\hat{T}) = H + o(1)$, so the mean-value theorem yields

$$|T_c - \hat{T}| = o(H^{-1}),$$

proving (42). $\square$

Proof of Theorem 1.7. The actual value $a_{n,k}$ can be written as $a(T_n)$ with T_n given by (6). Because $k = K_n + 1$ gives $a_{n,k} < B_k(\theta_n)$, we have $T_n \geq 0$. Let T_c be the exact center from Lemma 10.4. If $T = T_c + O(H^{-1})$, Taylor's theorem and (40)–(41) give

$$\Phi(T) = H + (1 + o(1))H(T - T_c) + O(H(T - T_c)^2),$$

where the quadratic term is $o(1)$. Hence

$$(44) \quad H - \Phi(T) = H(T_c - T) + o(1) = H(\hat{T} - T) + o(1),$$

using $T_c - \hat{T} = o(H^{-1})$.

If $H(\hat{T} - T_n)$ is bounded, then $T_n - T_c = O(H^{-1})$, so (44) applies. Conversely, suppose $H - \Phi(T_n)$ is bounded. Since the r_* term gives $\Phi(T_n) \geq Z(T_n) \geq e^{T_n}$, we have

$$T_n \leq \log(H + O(1)) = 2 \log k + O(1) < 3 \log k$$

for all large k . For every s between T_n and T_c , monotonicity gives $\Phi(s) = H + O(1)$. By (38), $Z(s) = \Theta(H)$, and then (40) yields $\Phi'(s) = \Theta(H)$ uniformly on this interval. The mean-value theorem therefore gives $T_n - T_c = O(H^{-1})$, so (44) applies again. This proves the stated equivalence.

Finally,

$$a(T) = B - \rho^{-r_*} T.$$

Thus a change of order H^{-1} in T corresponds to a change of order $(H\rho^{r_*})^{-1}$ in a . Since $L_* = B\rho^{r_*}$, the corresponding change in the relative deficit $1 - a/B$ is of order $(HL_*)^{-1}$. $\square$

10.4. Finite macroscopic bias. The exact tilted partition is needed in the full global range. When θk remains bounded, it reduces to the usual Gaussian prefactor.

Proposition 10.5. *Suppose $\theta k \rightarrow A \in [0, \infty)$. Let x_A be the maximizer in (4) and put*

$$g(A) = \frac{1}{x_A(1 - x_A)} + A^2 h(x_A).$$

Then

$$(45) \quad \Xi_{k,\theta} = (1 + o(1)) \sqrt{\frac{2\pi k}{g(A)}}.$$

Moreover, for every fixed $C_T > 0$, uniformly for $0 \leq T \leq C_T \log k$,

$$(46) \quad Z_{k,\theta}(T) = (1 + o(1)) e^T \Xi_{k,\theta}.$$

Consequently,

$$(47) \quad \hat{T} = \frac{3}{2} \log k + \log \log 2 - \frac{1}{2} \log \frac{2\pi}{g(A)} + o(1).$$

For $A = 0$, this becomes

$$\hat{T} = \frac{3}{2} \log k + \log \log 2 - \frac{1}{2} \log \frac{\pi}{2} + o(1).$$

Proof. Write $A_k = \theta k$, so $A_k \rightarrow A$, and let

$$f_k(r) = \log \binom{k}{r} - B_k(\theta) e^{\theta r} = \log u_r.$$

By [Lemma 6.5](#) and uniqueness of the maximizer of $x \mapsto h(x)e^{-Ax}$, every maximizing layer satisfies

$$\frac{r_*}{k} \rightarrow x_A.$$

In particular, both r_* and $k - r_*$ are of order k .

Put $d_r = f_k(r+1) - f_k(r)$. For $1 \leq r \leq k-1$,

$$\Delta^2 f_k(r) = \log \frac{r(k-r)}{(r+1)(k-r+1)} - B_k(\theta) e^{\theta(r-1)} (e^\theta - 1)^2.$$

Fix $M > 0$. Uniformly for $|r - r_*| \leq M\sqrt{k}$, we have $r/k = x_A + o(1)$. A uniform Taylor expansion of the two logarithms in the first term gives

$$\log \frac{r(k-r)}{(r+1)(k-r+1)} = -\frac{1}{kx_A(1-x_A)} + o(k^{-1}).$$

Moreover,

$$B_k(\theta) e^{\theta r_*} = \log \binom{k}{r_*} = kh(x_A) + o(k),$$

while $e^{\theta(r-r_*)} = 1 + o(1)$ uniformly in the window and $(e^\theta - 1)^2 = (A^2 + o(1))/k^2$. Hence

$$(48) \quad \Delta^2 f_k(r) = -\frac{g(A)}{k} + o(k^{-1})$$

uniformly for $|r - r_*| \leq M\sqrt{k}$.

Since r_* is a discrete maximum, $d_{r_*-1} \geq 0 \geq d_{r_*}$. As $d_{r_*} - d_{r_*-1} = \Delta^2 f_k(r_*)$, relation (48) gives $d_{r_*-1}, d_{r_*} = O(k^{-1})$. Summing the first differences, with the uniform error in (48), yields

$$f_k(r_* + m) = -\frac{g(A)}{2k} m^2 + o(1)$$

uniformly for $|m| \leq M\sqrt{k}$. Therefore, for fixed M ,

$$\frac{1}{\sqrt{k}} \sum_{|m| \leq M\sqrt{k}} e^{f_k(r_*+m)} \rightarrow \int_{-M}^M e^{-g(A)x^2/2} dx.$$

The global second-difference estimate from [Lemma 10.1](#) gives $f_k(r_* \pm m) \leq -cm(m-1)/k$ for an absolute $c > 0$. Consequently the remaining contribution, divided by $\sqrt{k}$, is bounded by a Gaussian tail that tends to zero as $M \rightarrow \infty$, uniformly in k . This proves (45).

Fix $C_T > 0$ and consider $0 \leq T \leq C_T \log k$. Choose $C_1 = C_1(C_T, A)$ sufficiently large and put $R = C_1 \sqrt{k \log k}$. On $|r - r_*| \leq R$,

$$\theta(r - r_*) = O\left(\sqrt{\frac{\log k}{k}}\right),$$

so

$$T(e^{\theta(r-r_*)} - 1) = o(1)$$

uniformly. Hence the summand in $e^{-T}Z(T)$ is $(1+o(1))u_r$ on this window. Outside the window, global concavity gives contact mass at most k^{-M_1} , where M_1 can be made arbitrarily large by increasing C_1 . Since A_k is bounded, the tilting factor is at most

$$\exp\{Te^{\theta k}\} \leq k^{C_T e^{A+1}}$$

for all large k . Choosing $M_1 > C_T e^{A+1} + 2$ makes the tilted tail $o(\Xi_{k,\theta})$. This proves (46).

Finally, Lemma 6.3 gives $H = (\log 2 + o(1))k^2$ and hence $\log H = 2 \log k + \log \log 2 + o(1)$. Solving $Z(\hat{T}) = H$ by (45) and (46) proves (47). At $A = 0$, $x_A = 1/2$ and $g(0) = 4$. $\square$

11. CONCLUDING REMARKS

The paper identifies one occupancy landscape behind three apparently different laws. At $p = 1/2$, the central Hamming layers contain a Gaussian number of comparable coupons and the standard coupon-collector correction appears. For fixed $p \neq 1/2$, finitely many rare layers control the threshold and produce the constant c_p . Between these endpoints the maximizer of $h(x)e^{-Ax}$ moves continuously from $1/2$ to zero, while the exact exponent $\Phi_{n,k}$ retains the lattice information needed for the critical Poisson law.

The fixed and global theorems use different convenient lattice parametrizations. At $p = 1/2$, the contact lattice K_n need not equal the classical J_n for every n , but the descriptions agree. Indeed,

$$B_k(0) = \log \binom{k}{\lfloor k/2 \rfloor} = k \log 2 - \frac{1}{2} \log k - \frac{1}{2} \log(\pi/2) + o(1),$$

so $K_n \in \{J_n, J_n + 1\}$. If $K_n = J_n$, the global critical level is exactly $J_n + 1$, and in the nondegenerate window $\Phi_{n,k} = 2^k F(u) = 2^k e^{-u} + o(1)$, recovering Theorem 1.1. If $K_n = J_n + 1$, the contact inequality gives $u \geq \log n - \frac{3}{2} \log \log n + O(1)$ at that level; hence the symmetric empty-box mean is $o(H_{n,k})$, and both formulations give $D_n = J_n + 1$ with high probability.

Two natural directions remain. The first is the complete sparse phase diagram when p_n tends to zero or one, where minimum cell means need not diverge and internal witnesses can be substantial. The second is the analogous theory for graph balls and distance VC-dimension. In dense inhomogeneous models, the fixed-state and continuous-kernel variational problems suggest further sharp lattice and crossover questions.

DECLARATION ON THE USE OF AI

The authors used generative AI tools to assist in discussing proof strategies, checking proofs, and improving exposition.

REFERENCES

- [1] M. Anthony, G. Brightwell, and C. Cooper. The Vapnik–Chervonenkis dimension of a random graph. *Discrete Mathematics*, 138(1–3):43–56, 1995.
- [2] A. D. Barbour, L. Holst, and S. Janson. *Poisson Approximation*. Oxford University Press, Oxford, 1992.
- [3] D. Coudert, M. Csikós, G. Ducoffe, and L. Viennot. Practical computation of graph VC-dimension. In *22nd International Symposium on Experimental Algorithms (SEA 2024)*, volume 301 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 8:1–8:20. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2024.
- [4] A. P. Godbole, D. E. Skipper, and R. A. Sunley. t -covering arrays: Upper bounds and Poisson approximations. *Combinatorics, Probability and Computing*, 5(2):105–117, 1996.
- [5] T. E. Harris. A lower bound for the critical probability in a certain percolation process. *Mathematical Proceedings of the Cambridge Philosophical Society*, 56(1):13–20, 1960.
- [6] E. Kranakis, D. Krizanc, B. Ruf, J. Urrutia, and G. Woeginger. The VC-dimension of set systems defined by graphs. *Discrete Applied Mathematics*, 77(3):237–257, 1997.

- [7] T. Pham, S. Senger, M. Tait, and N. Thu-Huyen. VC-dimension and pseudo-random graphs. *Discrete Applied Mathematics*, 365:231–246, 2025.
- [8] N. Sauer. On the density of families of sets. *Journal of Combinatorial Theory, Series A*, 13(1):145–147, 1972.
- [9] S. Shelah. A combinatorial problem; stability and order for models and theories in infinitary languages. *Pacific Journal of Mathematics*, 41(1):247–261, 1972.
- [10] V. N. Vapnik and A. Y. Chervonenkis. On the uniform convergence of relative frequencies of events to their probabilities. *Theory of Probability and its Applications*, 16(2):264–280, 1971.
- [11] B. Ycart and J. Ratsaby. The VC dimension of k -uniform random hypergraphs. *Random Structures & Algorithms*, 30(4):564–572, 2007.
- [12] B. Ycart and J. Ratsaby. VC-dimensions of random function classes. *Discrete Mathematics & Theoretical Computer Science*, 10(1):113–128, 2008.

SCHOOL OF MATHEMATICAL SCIENCES, UNIVERSITY OF SCIENCE AND TECHNOLOGY OF CHINA, HEFEI, CHINA
Email address: liuxizhi@ustc.edu.cn

SCHOOL OF MATHEMATICS, SHANDONG UNIVERSITY, JINAN, CHINA; AND EXTREMAL COMBINATORICS AND PROBABILITY GROUP, INSTITUTE FOR BASIC SCIENCE, DAEJEON, SOUTH KOREA
Email address: heng.li@sdu.edu.cn